

Cisco Networking Basic Summary



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Module(1)

1. Network Types – The Many Faces of Connectivity

What is the Internet?

- The **Internet** is a **global network of networks** that connects billions of devices worldwide.
- It is **not owned by anyone** – it's a cooperative system that uses common standards to share information.
- We use it daily for: social media, shopping, learning, gaming, messaging, and more.

Who Owns the Internet?

- **No single person or company owns it.**
- It works through **telephone lines, fiber optics, wireless signals, and satellites.**
- All online services (like websites, games, and emails) are hosted on local networks that connect to the internet.

2. Different Sizes of Networks

Network Type	Description	Examples
Small Home Network	Connects a few devices at home to the internet	Home Wi-Fi, shared printer
SOHO (Small Office/Home Office)	Small business or home office networks	Connecting laptops, printers, and files locally
Medium to Large Networks	Used by businesses for operations, sales, and customer service	Corporate offices, data centers
World Wide Networks	The entire global internet	All connected local and wide-area networks

3. Mobile & Connected Devices

Mobile Devices

- **Smartphones, Tablets, Smartwatches, Smart Glasses**
- Can access the internet from almost anywhere.
- Combine multiple functions: calling, cameras, GPS, media, apps.

Connected Home Devices

- Devices in your home connected to the internet for remote control.
 - Security Systems
 - Smart TVs
 - Gaming Consoles
 - Smart Appliances & Lighting

Other Connected Devices (Outside Home)

- **Smart Cars:** Connect to maps, entertainment, diagnostics, and even emergency services.
- **RFID Tags & Sensors:** Used for tracking packages, monitoring environments.
- **Medical Devices:** Connected health monitors, diagnostic tools.





4. What is a Network? (Quick Overview)

- A **network** is simply **two or more devices connected together** to share resources or information.
 - Can be **wired (cables)** or **wireless (Wi-Fi, Bluetooth, cellular)**.
 - Allows for **communication, resource sharing (printers, files), and internet access**.
-



5. Data Transmission – How Data Moves



The Basics: Bits & Bytes

- Computers only understand **binary** – **0s and 1s** (called **bits**).
- **8 bits = 1 Byte**
- **ASCII Code** is used to represent letters, numbers, and symbols in binary.



How is Data Sent Over Networks?

Data is turned into **signals** and sent through a **medium (physical or wireless)**:

1. **Electrical Signals** – Data sent as electric pulses over **copper wires**.
2. **Optical Signals** – Data sent as **light pulses** through **fiber-optic cables**.
3. **Wireless Signals** – Data sent via **radio waves, microwaves, or infrared** (Wi-Fi, Cellular).



6. Bandwidth & Throughput – How Fast is Your Connection?



Bandwidth

- **Definition:** The **maximum capacity** of a network connection to carry data.
- Think of it as the **size of the “pipe”** – how much data can flow at once.
- Measured in **bits per second (bps)**, Mbps, Gbps.



Throughput

- **Definition:** The **actual amount of data** successfully transferred in a given time.
- Usually slower than bandwidth due to:
 - Network congestion
 - Distance
 - Number of devices
 - Latency (delays)
- **Latency** = Delay in data travel time from source to destination.

⚠ **Important:** Even if you have high bandwidth, your **throughput can be low** if there are delays or weak links in the network.

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	Kbps	1 Kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

? Check Your Understanding – Simplified Q&A

Network Types

1. What allows remote workers to connect to their office network?
 - ☒ Small Office / Home Office (SOHO) Network
 2. What helps track packages?
 - ☒ RFID Tag
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Data Transmission

1. Smallest unit of data?
 - ☒ Bit
 2. Data collected about you without you directly giving it?
 - ☒ Inferred Data
 3. Signal sent using light?
 - ☒ Optical Signals
-

Bandwidth & Throughput

1. Capacity to carry data?
 - ☒ Bandwidth
2. Measurement that includes delays?
 - ☒ Throughput
3. Unit that includes latency effects?
 - ☒ Bits per second / Bytes per second

Key Takeaways – Easy to Remember

Topic	Key Point
Internet	A global, shared network owned by no one, used by everyone.
Networks	Allow devices to share data, resources, and internet access.
Devices	Phones, watches, appliances, cars – all can be smart and connected.
Data	Computers use bits (0s & 1s) to process everything.
Signals	Data travels as electricity, light, or radio waves .
Bandwidth	The potential speed of your connection.
Throughput	The actual speed you experience, including delays.

Bonus: Why This Matters

- Understanding these basics helps you:
 - Troubleshoot network issues
 - Choose better internet plans
 - Build or manage networks (at home or work)

Summary in One Line:

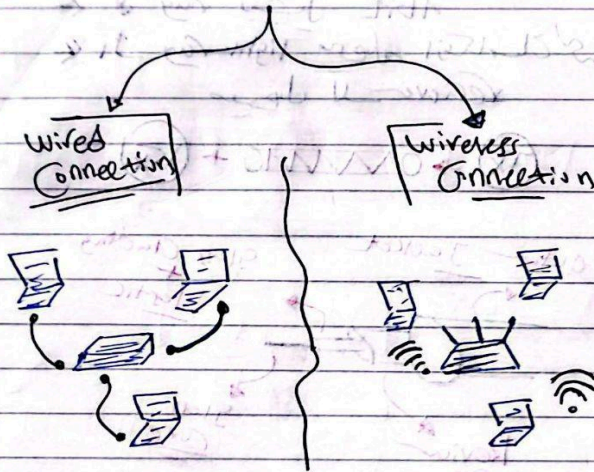
Networks connect devices using signals (wired or wireless) to share data, and how fast/smart that connection is depends on bandwidth, throughput, and the type of data being sent.

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Network Course (IT_Duse11)

Network?

مجموعة من الأجهزة المتصلة ببعضها البعض



Uses of Network ---

1) Sharing Hardware

يمكن لأجهزة على شبكة واحدة مشاركة أجهزة أخرى مثل طابعات أو كاميرات أو أجهزة تخزين. في شبكة يمكن مشاركة أجهزة مثل طابعات أو أجهزة تخزين. في شبكة يمكن مشاركة أجهزة مثل طابعات أو أجهزة تخزين. في شبكة يمكن مشاركة أجهزة مثل طابعات أو أجهزة تخزين.

2) Sharing data

يمكن لأجهزة على شبكة واحدة مشاركة البيانات مثل ملفات أو صور أو مقاطع فيديو أو برامج. في شبكة يمكن مشاركة البيانات مثل ملفات أو صور أو مقاطع فيديو أو برامج. في شبكة يمكن مشاركة البيانات مثل ملفات أو صور أو مقاطع فيديو أو برامج.

* Time ↓
Performance ↑

(3) data Protection

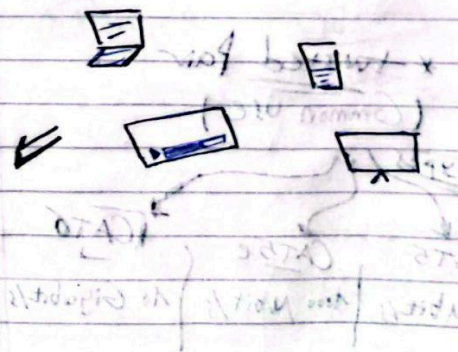
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Network Components

- 1) End devices
- 2) Network media
- 3) Network interface
- 4) Network protocol

1) End devices

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Module(2)

1. Clients and Servers – Who Does What on the Network?

Hosts:

Any device connected to a network (like a PC, phone, or printer) that can **send or receive data** is called a **host**.

Two Main Roles:

Role	What It Does	Example
Client	Requests services or data (e.g., opening a website)	Your laptop when you browse the internet
Server	Provides services or data (e.g., hosts a website)	A computer that stores and shares files or websites

 The **software** on the device decides if it's acting as a **client, server, or both**.

2. Peer-to-Peer (P2P) Networks – Everyone Shares

What Is It?

- In a **P2P network**, every device can be both a client and a server.
- No central server is needed. Devices **share files, printers, or services directly with each other**.

Where It's Used:

- **Homes and small offices** for simple tasks.
- Only needs **basic connections** like Wi-Fi or cables.

How It Works:

- Two or more computers are connected (wired or wireless).
- One can **share a printer**, the other can **access files**.
- A **switch** may be used to connect more devices.

Advantages:

- **Easy to set up**
- **Low cost** (no special servers needed)
- **Good for simple tasks** like file sharing or printing

Disadvantages:

- **No central control** (harder to manage security)
- **Not secure**
- **Not ideal for large networks**
- **Slower performance** if a device is both giving and requesting data



3. Peer-to-Peer Applications – Everyone is a Client & Server

- Apps where **each device gives and gets data at the same time**.
 - Example: **Chat apps** where both users can send and receive messages.
 - Some apps use a hybrid system:
 - Resources are shared between peers (decentralized)
 - But a central index helps locate the resources
-



4. Devices Can Have Multiple Roles

- One computer can:
 - Be a file server, web server, and email server at the same time
 - Also run multiple client apps like a web browser, email app, and messaging app



You need the **right software (clients)** to connect to different types of servers.



5. Network Infrastructure – The Foundation of All Networks



What Is It?

- The **physical and logical parts** that allow devices to **communicate**.
- Includes everything that makes networking possible.

Three Main Parts:

Category	What It Means	Examples
End Devices	Devices people use	Computers, phones, printers, tablets
Intermediate Devices	Devices that move/connect data	Routers, switches, firewalls, access points
Network Media	How data travels	Cables (Ethernet), Wi-Fi, fiber optics, air (wireless)

6. End Devices – The User’s Tools

These are the devices **you directly use** to access the network.

Common Examples:

- **Computers** (laptops, desktops, servers)
- **Printers**
- **Phones & Tablets**
- **Cameras & Video Phones**
- **Scanners & Barcode Readers**

 Each device has a **unique address** so data knows where to go.

7. How Do We Connect to the Internet? (ISP Basics)

What's an ISP?

- **Internet Service Provider** – the company that connects your home or office to the internet.
- Examples: we, Comcast, AT&T, Verizon, local telecom/cable companies

What Do ISPs Offer?

- Internet access
 - **Email & website hosting**
 - **Cloud storage**
 - **Security & backup services**
 - **Phone services (VoIP)**
-

8. How Do Devices Connect to an ISP?

Basic Option (Not Safe):

- Just a **modem connected directly to a computer** (no protection)

Recommended Option:

- A **router (often with Wi-Fi)** connects multiple devices safely to the ISP.
- The router gives **private IP addresses** and adds **security (like a firewall)**.



9. Common Ways to Connect to the Internet

Type	How It Works	Best For
Cable	Uses the same coaxial cable as cable TV	Fast, always-on internet
DSL	Uses telephone lines (voice + data together)	Good speed, available in many areas
Fiber-Optic	Uses thin glass fibers & light (very fast!)	Best performance, used in cities
Cellular	Uses mobile phone networks	Good for remote/mobile users
Satellite	Uses satellites in space	For rural areas with no other options
Dial-Up	Old method using phone lines & modems	Very slow, hardly used today

10. Cable vs. DSL – Home Internet Favorites

Cable Internet:

- Delivered over **coaxial cable** (same as cable TV)
- **High speed, always on**
- Uses a **cable modem**

DSL Internet:

- Delivered over **phone lines**
 - **Voice and internet** can work at the same time
 - Has three channels:
 1. Voice (phone calls)
 2. Download (fast incoming)
 3. Upload (slower outgoing)
 - Speed depends on **distance from phone company's hub**
-

11. Other Internet Options

Type	Details
Cellular	Use mobile networks anywhere you have signal. Good for travelers. May have data limits.
Satellite	Good for rural areas. Needs a clear view of the sky. Slower and more expensive.
Dial-Up	Ancient tech, very slow. Only for emergencies or areas with no other options.

Quick Recap – Key Points

Topic	What You Should Know
Clients & Servers	Clients request, servers provide. Software decides the role.
P2P Networks	All devices share equally. Good for small/simple networks.
P2P Apps	Every device is both client & server (like chat apps).
End Devices	The gadgets we use (PCs, phones, printers).
Network Infrastructure	Includes devices (routers, switches) and media (cables, Wi-Fi).
ISPs	Give your home or office access to the internet.
Internet Connections	Choose based on speed, location, and budget: Cable, DSL, Fiber, Cellular, Satellite, or Dial-up.

? Common Questions

1. What is the difference between a client and a server?

- A **client** asks for data/services, and a **server** provides them.

2. Can one device be both a client and a server?

- Yes, especially in **P2P networks** or when using multiple apps.

3. What is a P2P network good for?

- Small networks for **file sharing, printing, or simple tasks**.

4. What's the most common way to get internet at home?

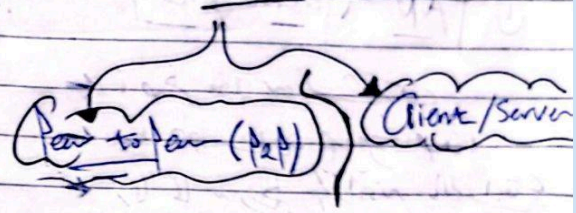
- **Cable or DSL** using a **router/modem combo**.

5. What if I live in a rural area?

- Try **satellite or cellular internet** as alternatives.

Types of Network

Host Role

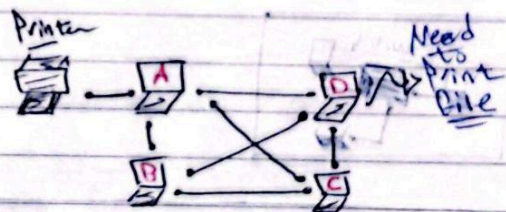


Client → Request Service
 (Computer / Phone)

Server → Receive Client Service
 (Provide Service)

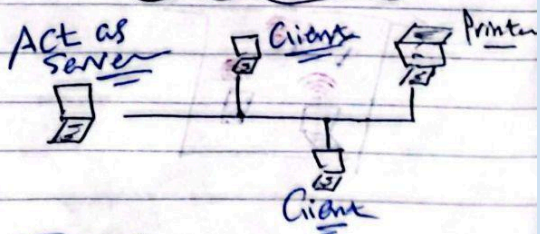
Peer-to-Peer

No Centralized device
 Server & Client are same
 All are Client & all are Server



Node A is Client & Node D is Server
 Printer is also A & A is Service
 Server is A & A is

Client/Server



Printer is also Server & A is Service
 2006/11

Module(3)

1. Wireless Networks (No Wires!)

These are networks that let devices connect and share data **without using cables**. Examples include:

Wi-Fi (Wireless Fidelity)

- Connects your phone, laptop, or tablet to the **internet or local network** using a **wireless router**.
- You need to be **within range** of a Wi-Fi signal (like at home, school, or café).
- **Public Wi-Fi hotspots** are available in many public places.
- Tips for safety:
 - Don't enter passwords on unsecured Wi-Fi.
 - Use WPA2 or higher encryption for your home Wi-Fi.
 - Consider using a VPN when sending private info.

Cellular/Mobile Networks (4G, 5G, GSM)

- Provided by your mobile carrier (like Vodafone, AT&T, etc.).
- Lets you access the internet **from almost anywhere** using mobile towers.
- Common types: **4G, 5G, 3G** – faster connections mean better streaming and downloads.
- **Uses data plans**, which may have limits or costs depending on your subscription.
- Your phone **automatically switches** between stronger signals (like from 4G to 3G) without you noticing.

GPS (Global Positioning System)

- Uses **satellites** to find your **exact location** (like in maps and navigation apps).
- Super accurate (within about 10 meters).
- Works **without internet**, but needs a clear view of the sky.

✓ Bluetooth

- A **short-range wireless technology** (works within a few meters).
- Used to connect devices without cables, like:
 - Headphones or speakers
 - Keyboards & mice
 - Car stereos or hands-free calling
 - Smartwatches
 - Portable speakers
- **Very low power**, so it helps save your device's battery.
- **Up to 8 devices** can be connected at the same time.

✓ NFC (Near Field Communication)

- Works **only when devices are very close** (a few centimeters apart).
 - Used for things like:
 - Contactless payments (like Apple Pay or Google Pay)
 - Tapping your phone to a transit card reader
 - Sharing files between phones quickly
-

2. How Mobile Devices Connect (Phones & Tablets)

Wi-Fi on Mobile Devices

- Most phones can connect to **Wi-Fi networks** just like laptops.
- Why use Wi-Fi?
 - It's usually faster than mobile data.
 - Doesn't use your mobile data plan.
 - Saves battery compared to cellular radios.
- How to connect:
 - Turn on Wi-Fi in your phone's settings.
 - Your phone will show a list of nearby Wi-Fi networks.
 - Tap the network name and enter the password (if needed).
 - Your phone can remember networks and reconnect automatically.

Cellular Data Settings

- Mobile data lets you go online **when there's no Wi-Fi**.
 - You can turn cellular data on or off in your phone's settings:
 - Android: Settings > More > Mobile Networks > Data Enabled
 - iPhone (iOS): Settings > Cellular Data (toggle on/off)
 - Your phone will **automatically use Wi-Fi if available**, and switch to mobile data if not.
-

3. Bluetooth – Easy Wireless Connections

What Can You Do With Bluetooth?

- Connect **wireless headphones or earphones**
- Use a **wireless keyboard or mouse**
- Play music through your **car stereo or speakers**
- Answer calls with a **hands-free Bluetooth device**
- **Share your phone's internet** (called tethering) with another device
- Connect to **Bluetooth speakers** for better sound

Bluetooth Pairing (Connecting Devices)

- **Step 1:** Turn on **Bluetooth** on both devices.
- **Step 2:** Put one device in **“Discoverable Mode”** (so the other can see it).
- **Step 3:** Search for devices from one phone/device.
- **Step 4:** Select the device you want to connect to.
- **Step 5:** Enter a **PIN** (sometimes required) to pair.
- **Done:** Once paired, they'll **remember each other** and connect automatically next time!

 **Tip:** On Android & iPhone, the steps are slightly different, but both are easy. Check your device's **Settings > Bluetooth** section.

Quick Tips to Remember

Technology	Range	Use Cases	Power Use	Safety Tips
Wi-Fi	Medium (Home/Office)	Internet, streaming, downloads	Medium	Use WPA2, avoid public unprotected Wi-Fi
Bluetooth	Short (Few meters)	Headphones, speakers, accessories	Low	Pair only with trusted devices
Cellular (4G/5G)	Wide (City/Country)	Internet anywhere, calls, messages	High	Watch your data usage
GPS	Global (Satellite)	Location, maps, navigation	Low	Needs open sky for best signal
NFC	Very Short (Few cm)	Payments, quick sharing	Very Low	Only tap trusted devices

Summary in Simple Words:

- **Wi-Fi** = Connects to the internet at home, school, or café (use it to save mobile data & battery).
- **Bluetooth** = Connects headphones, speakers, or accessories wirelessly (great for hands-free use).
- **Cellular Data** = Lets your phone get online **anywhere** using mobile signals (uses your data plan).
- **GPS** = Helps your phone know **exactly where you are** (for maps and navigation).
- **NFC** = Lets you **tap your phone** to pay or share info **super quickly**.

Module(4)

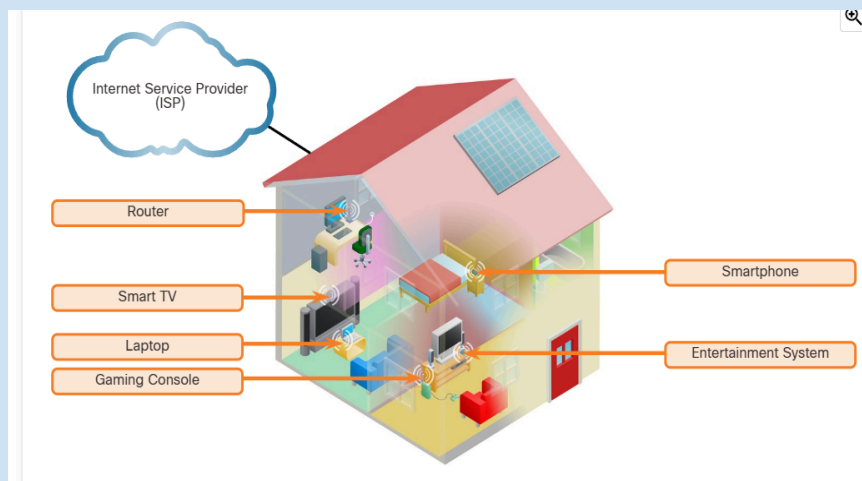
1. What is a Home Network?

A **home network** lets all your devices — like phones, laptops, TVs, and game consoles — **connect to each other and to the internet.**

Common Devices in a Home Network:

- **Computers** (laptops, desktops)
- **Gaming consoles** (PlayStation, Xbox, etc.)
- **Smart TVs**
- **Printers & Scanners**
- **Smartphones & Tablets**
- **Security cameras**
- **Climate control devices** (like smart thermostats)
- **Telephones (VoIP)**

 These devices connect either **wirelessly (Wi-Fi)** or **through cables (Ethernet)** to a **central router**.





2. Components of a Home Network



Main Device: The Router

- The **router** is the **heart of your home network**.
- It connects all your devices to **each other** and to the **internet**.



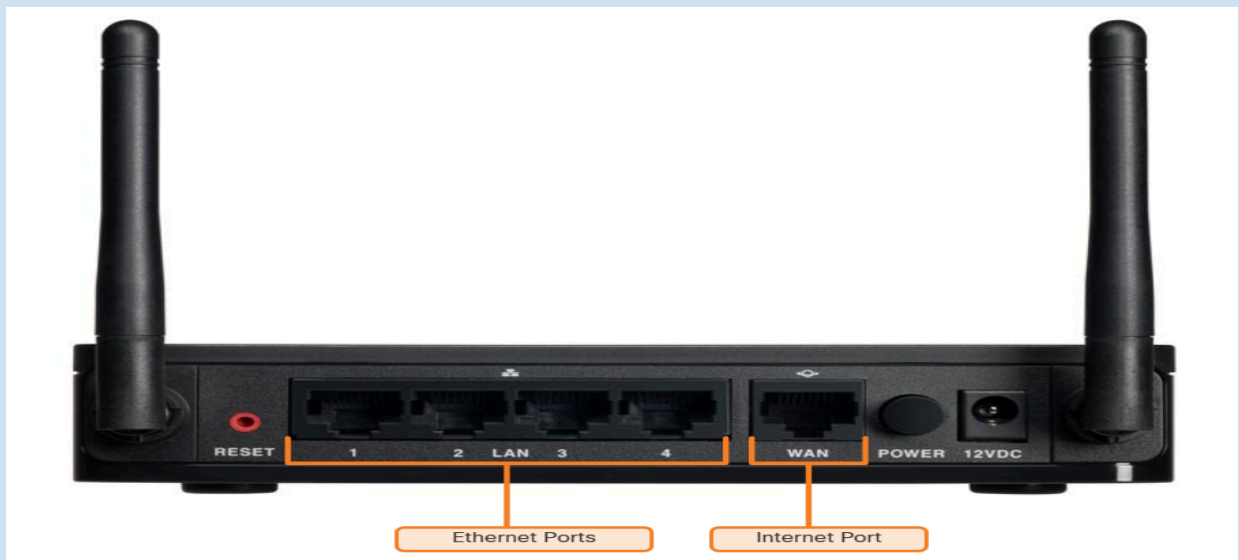
Router Ports:

1. Ethernet Ports (LAN Ports):
 - Used to connect wired devices like computers or gaming consoles.
 - All devices plugged here are on the same local network.
2. Internet Port (WAN Port):
 - Connects the router to the modem (which connects to your Internet Service Provider or ISP).
 - This is what gives your home access to the internet.



Most routers also have:

- **Wi-Fi antennas** → for wireless connections
- **Built-in wireless access point** → so phones and laptops can connect without cables





3. Wireless Technologies in the Home



Wi-Fi (Wireless LAN)

- Uses **radio waves** to connect devices **without cables**.
- Based on **IEEE 802.11 standards** (like 802.11b, g, n, ac, ax → known as **Wi-Fi**).
- Works on two main frequency bands:
 - 2.4 GHz → longer range, but slower and more crowded
 - 5 GHz → faster speeds, but shorter range



Bluetooth

- Also uses the **2.4 GHz band**.
 - Used for short-range connections, like:
 - Headphones
 - Keyboards & mice
 - Speakers
 - Smartwatches
-



4. Wired Network Technologies

Even though Wi-Fi is common, **wired connections** are still useful for **faster, more stable** connections.



Ethernet (Wired LAN)

- Uses **Ethernet cables (RJ-45)** to connect devices directly to the router or switch.
- Most common cable: **Category 5e (Cat 5e)**
- Other options (less common in homes):
 - Coaxial cable (used for TV/internet)
 - Fiber-optic cable (very fast, less common in houses)
 - Powerline adapters (use your home's electrical wiring for internet)



5. Wi-Fi Standards & Settings



Wi-Fi Standards (IEEE 802.11)

- Created by the **Institute of Electrical and Electronics Engineers (IEEE)**.
- Govern how Wi-Fi devices **communicate, their speed, and range**.
- The **Wi-Fi Alliance** tests devices to make sure they work well together.



Common Wi-Fi Router Settings:

1. Network Mode
 - Decides which Wi-Fi standards (like 802.11b/g/n/ac) the router will support.
 - Mixed Mode = supports both old and new devices.
2. SSID (Network Name)
 - This is the name of your Wi-Fi network that appears on your phone or laptop.
 - Don't use personal info (like your name or address) in the SSID.
 - SSID Broadcast = lets devices see your network name automatically.
3. Channel
 - Determines which radio channel your Wi-Fi uses (usually set to Auto).



Security Tip: Always use strong **Wi-Fi passwords** and the latest **encryption** (like **WPA2** or **WPA3**) to protect your network.

✓ Quick Summary Checklist:

Topic	Key Points
Home Network	Connects all your devices (phones, laptops, TVs, etc.) to each other and the internet
Router	The main device that manages your home network and internet connection
Ports	LAN (Ethernet) ports = wired devices, Internet/WAN port = connects to modem & internet
Wi-Fi	Wireless connection using 2.4 GHz or 5 GHz , based on IEEE 802.11 standards
Bluetooth	Short-range wireless for accessories like headphones, mice, speakers
Ethernet	Wired connection using Ethernet cables (RJ-45) for stability & speed
Router Setup	Connect via cable, use default Wi-Fi name/password, then secure it properly
Security	Always use WPA2/WPA3 encryption , a strong password , and consider a guest network

Module(5)

1. Why Communication Rules (Protocols) Matter

Human Communication Analogy:

- When two people talk, they need:
 - A common language (like English)
 - An agreed method (talking face-to-face, texting, etc.)
 - To confirm understanding (like nodding or signing a paper)

 The same rules apply to computers and networks.

Network Communication Needs:

- **Identified sender and receiver**
- **Agreed communication method** (like Wi-Fi, cables)
- **Common language (protocol)** so devices understand each other
- **Speed and timing** of sending/receiving data
- **Confirmation** that messages were received (like acknowledgments)

 These rules are called **protocols**, and they ensure successful communication between devices.

2. Communication Standards – Making Devices Work Together

What Are Standards?

- **Standards are agreed-upon rules** that make sure **all devices** (computers, phones, printers, etc.) can **communicate properly** over the internet or a network.

Example:

- Email works the same on **phones, tablets, and computers** because they all follow the **same email standards**.
- Without standards, devices **wouldn't understand each other**.

Who Creates These Standards?

- Managed by special organizations that test, develop, and publish rules.
- One key group is the **Internet Engineering Task Force (IETF)**.
- They write detailed documents called **RFCs (Request for Comments)** to track how standards are created and improved.

 **Standards ensure compatibility, reliability, and global communication between all kinds of devices.**

3. Network Communication Models – How Data Travels

These are like **blueprints or layered maps** that show **how data moves from one device to another** across networks.

There are **two main models**:

A. The OSI Model (Open Systems Interconnection)

- Developed by the **International Organization for Standardization (ISO)**
- Has **7 layers** that describe **how data is prepared, sent, and received**
- Each layer has a specific job, like:
 - Formatting the data
 - Adding addresses
 - Routing through networks
 - Ensuring delivery

The 7 OSI Layers (from top to bottom):

1. **Application** (user software like browsers)
2. **Presentation** (data translation, encryption)
3. **Session** (managing connections)
4. **Transport** (reliable data delivery)
5. **Network** (routing data)
6. **Data Link** (error detection, MAC addresses)
7. **Physical** (cables, signals, hardware)

B. The TCP/IP Model (Transmission Control Protocol / Internet Protocol)

- The **model actually used on the real internet today**
- Has **4 (or sometimes 5) layers** that combine some of the OSI layers
- Focuses on **how the internet really works**
- Includes real protocols like **TCP, IP, HTTP, DNS, etc.**

The 4 TCP/IP Layers:

1. **Application** (like browsers, email)
2. **Transport** (like TCP – ensures data arrives safely)
3. **Internet** (like IP – routes data across networks)
4. **Network Access (or Link)** (cables, Wi-Fi, physical connections)

 **The TCP/IP model is what the internet actually uses. The OSI model is a theoretical guide that helps understand all the layers involved.**

4. Comparing OSI & TCP/IP Models

Feature	OSI Model	TCP/IP Model
Number of Layers	7 layers	4 (or 5) layers
Usage	Theoretical/model for learning	Actual model used on the Internet
Layer Grouping	Separates Physical & Data Link	Combines them into "Network Access"
Flexibility	More detailed, good for teaching	Simpler, used in real-world networking
Mapping	OSI Layer 3 = TCP/IP Internet Layer OSI Layer 4 = TCP/IP Transport Layer	Same core functions, different layer names

 **Both models are used together** when talking about how devices and apps communicate over networks.

✓ Quick Summary – Key Points to Remember

Topic	What You Should Know
Communication Protocols	Rules that devices follow to talk to each other (like language & method in human chat)
Why Protocols Exist	To ensure messages are sent, received, and understood correctly
Standards	Agreed rules that let different devices work together (like email, web pages, videos)
Who Makes Standards?	Organizations like IETF that publish rules as RFCs
OSI Model	7-layer theory model that explains ALL steps of data communication
TCP/IP Model	4-layer practical model used by the real internet
Key Similarities	Both models define how data moves between devices in layers
Key Differences	OSI is more detailed and theoretical; TCP/IP is simpler and used in real life

IEEE

OSI Model (Open System Inter Connection)

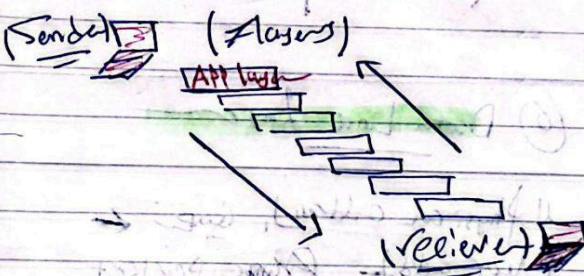
Network (Network Standard)
Operat Syst

IEEE
IEEE

IEEE

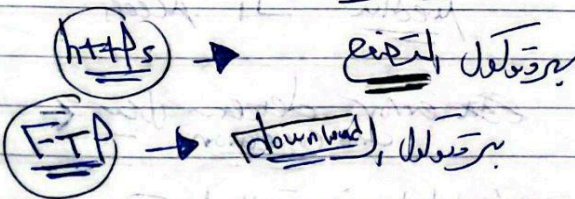
IEEE

IEEE



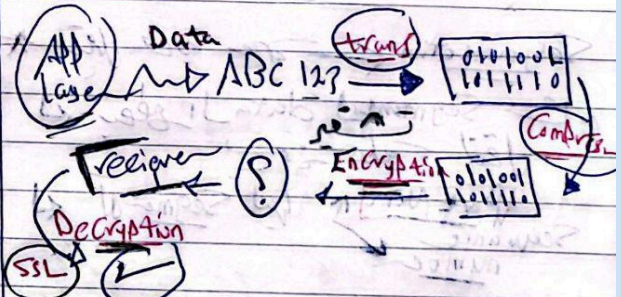
Application layer

Protocol
Action



Presentation Layer

Machine lang



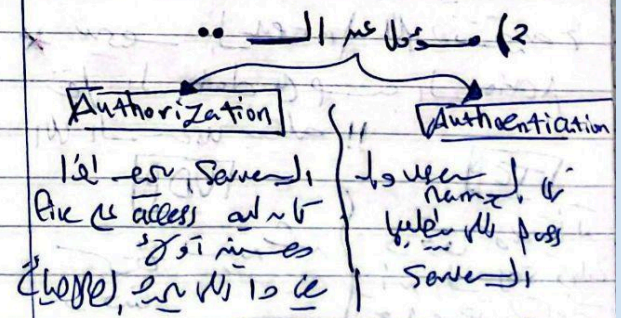
Machine lang

Machine lang

Machine lang

Session Layer

Session

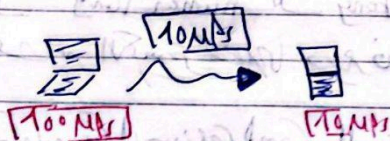


Session Management

(4) Transport Layer

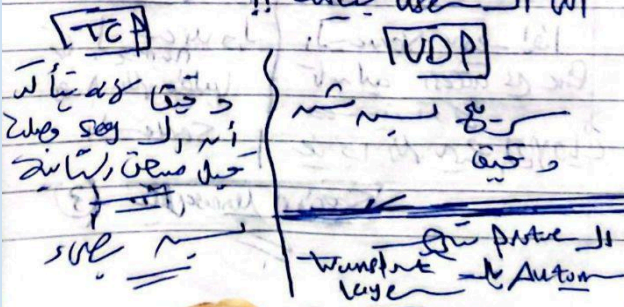
data transfer *
 Segmentation → data is divided into segments *
 Port Num. Sequence number

Flow Control, no loss *
 transfer rate, ie



لا يزال يعمل بمرور الوقت و...
 data loss, error, and...
 performance

Transmission Control Protocol (TCP) *
 Action, data, use



(5) Network Layer

IP, logical Address, Routing, packet (data unit)

(6) Data Link Layer

Physical address, MAC address, frame

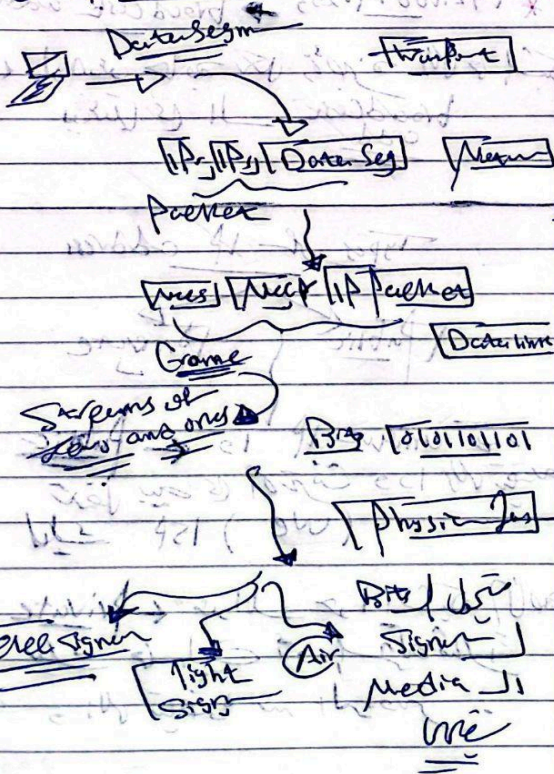
upper layers, media access, error detection and correction

Parity Checking

Checksum → 0: technique for error detection
 CRC → error detection
 data error in frame, not in data
 data is discarded

CMA → 2nd Protocol for IP traffic
 Media Cable
 data is not lost

Physical Layer



rel (3)

<< IP address >>

* عنوان هر کامپیوتر *
 در شبکه باید یکتا باشد
 - 0 تا 255

(ex) 192.168.1.6

8 bit → 0 تا 255
 (0, ..., 255)

IP → 4 octet

① Network address

② Host address

» « 192.168.1.6 »

Subnet Mask

IP → 32 bit
 Network address → 1st 3 octets
 Host address → last octet

(ex) IP add: 192.168.1.6
 S.M: 255.255.255.0

Network address → 1st 3 octets
 Host address → last octet

Module(6)



1. What is Network Media?

- **Network media** is the **physical or wireless path** that **data travels through** from a **source device (like a computer)** to a **destination device (like a server or printer)**.
 - Think of it as the **"highway"** that your data drives on to get from one place to another.
-



2. Three Main Types of Network Media

Modern networks use **three main types of media** to send data:



1. Metal Wires (Copper Cables)

- Data is sent as **electrical signals** through metal wires inside cables.
- Examples: **Twisted-pair cables** and **coaxial cables**



2. Fiber-Optic Cable (Glass or Plastic Fibers)

- Data is sent as **pulses of light** through **thin strands of glass or plastic**.
- **Super fast and reliable**, especially for long distances.



3. Wireless Transmission

- Data is sent through the **air** using **invisible electromagnetic waves** (like radio or Wi-Fi signals).
- No physical cables needed — used in Wi-Fi, Bluetooth, and cellular networks.

3. Four Key Factors When Choosing Network Media

When deciding which type of media to use, network engineers think about:

1.  Distance
 - How far the signal can travel without getting weak or lost?
 2.  Environment
 - Where will the cable or signal be? (e.g., indoors, outdoors, underground, etc.)
 3.  Speed & Bandwidth Needs
 - How much data needs to be sent, and how fast?
 4.  Cost
 - How expensive is the media, and how much will installation cost?
-

4. Common Types of Network Cables

Here are the **three most common types of wired network cables**:

1. Twisted-Pair Cable (Most Common)

- **What it is:** Copper wires twisted together in pairs to **reduce interference**.
- **Used for:** Most **Ethernet networks** (like in homes, schools, and offices).
- **Looks like:** Ethernet cables (often with **RJ-45 connectors**).
- **Colors:** Each wire pair has a solid color and a striped version (for easy matching).

✓ Pros: Cheap, flexible, easy to install

✗ Cons: Slower and more interference-prone than fiber

2. Coaxial Cable

- **What it is:** A **single copper core** surrounded by insulation, metal shielding, and a protective jacket.
- **Used for:** Older TV cable systems, some satellite connections, and early networks.
- **Not commonly used for modern computer networks anymore.**

✓ Pros: Good for carrying high-frequency signals

✗ Cons: Bulky, not ideal for high-speed data networks today

3. Fiber-Optic Cable

- **What it is:** Thin strands of **glass or plastic** that carry data as **light pulses**.
- **Used for:** High-speed internet, long-distance communication, and backbone networks.
- **Superior speed & distance**, and **not affected by electrical interference**.

✓ Pros: Extremely fast, long range, secure, no interference

✗ Cons: More expensive, harder to install and repair

5. Wireless Media (No Cables!)

- **What it is:** Data travels through the **air** using **radio waves, microwaves, or infrared signals**.
- **Examples:** **Wi-Fi, Bluetooth, Cellular (4G/5G), Satellite**
- **Used for:** Mobile devices, wireless routers, smart homes, etc.

✓ Pros: No cables, flexible, great for mobility

✗ Cons: Can be slower, affected by distance/walls/interference

✓ Quick Summary Table: Network Media Types

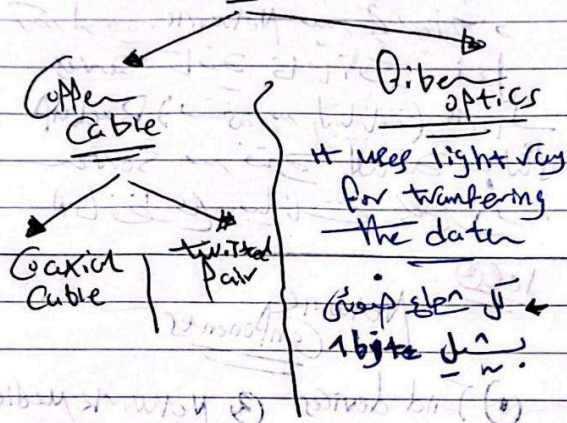
Media Type	How It Works	Speed	Distance	Cost	Best For
Twisted-Pair (Ethernet Cable)	Electrical signals over copper wires	Medium	Short–Medium	Low	Home/Office Networks
Coaxial Cable	Electrical signals over a copper core	Medium	Medium	Medium	Cable TV, Legacy Networks
Fiber-Optic Cable	Light pulses through glass/plastic fibers	Very High	Long	High	High-Speed Internet, Long Distances
Wireless (Wi-Fi, Bluetooth, etc.)	Radio waves through the air	Medium–High	Medium–Long	Medium	Mobile Devices, No-Wires Setup

[2] Network Media (Connection)

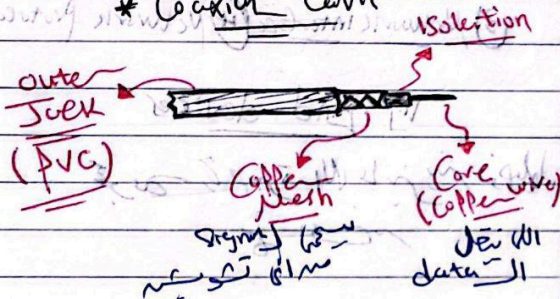
- * Wires (cables)

- * wirel (radio free)

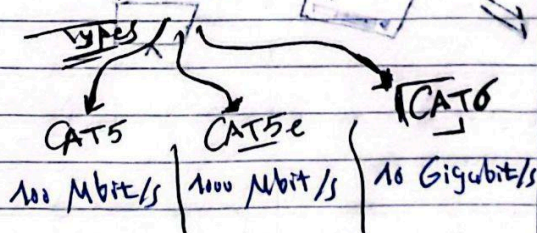
Cables



* Coaxial Cable



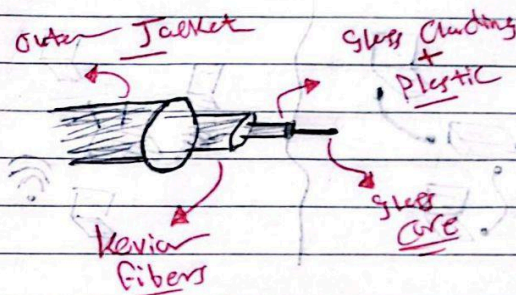
* twisted Pair
(Common use)



celler, after ←
cable
de 100m ne el, el el ni
daten ip gross case

* Fiber optics *

light ray کی طرف سے ←
 1 bit کی طرف سے ←
 where light ray کی طرف سے ←
 relieve کی طرف سے ←



[3] Network interface

Sut exp exp data J low

the re

if pc → pc m data bus k f d *

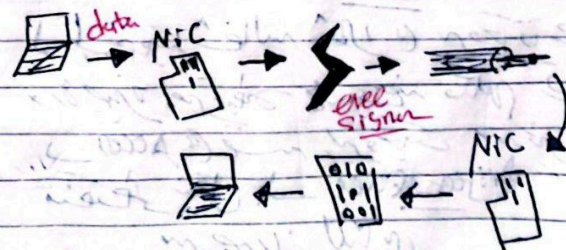
Copper Cable p s i n t

m Blue to b l e n W data bus *

Copper Cable → .. -- (ones + zeros)

Junk bits .. elec signal r s

elec Sign ← (0,1) m data d



* Light Senses (درد و حس نور)

Module(7)

1. What is Encapsulation? (Like Putting a Letter in an Envelope)

Real-Life Example: Sending a Letter

- When you send a letter, you don't just write your message on a blank paper and mail it.
- You put it inside an envelope, and write:
 - Your address (sender)
 - Recipient's address (destination)
 - And then seal it to be delivered properly.

 The same idea applies to computer messages (data) sent over a network.

In Networking:

- Your message (like an email or file) is **wrapped or packaged** inside a **special format called a "frame"**.
- This process is called **encapsulation**.
- The frame adds important details like:
 - Source & destination MAC addresses (like sender & receiver addresses)
 - Control information (like stamps or labels on an envelope)
- At the receiving end, this wrapping is **removed in a process called de-encapsulation**, so the original message can be read.

 **Why it matters:** If data isn't properly packaged (encapsulated), it **can't be delivered or understood** by the receiving device.



2. What is an Ethernet Frame?



Ethernet = The Most Common Wired Network Technology

- Used in **local networks (LANs)**, like in homes, schools, or offices.
- Devices like **computers, printers, and switches** use **Ethernet** to communicate.

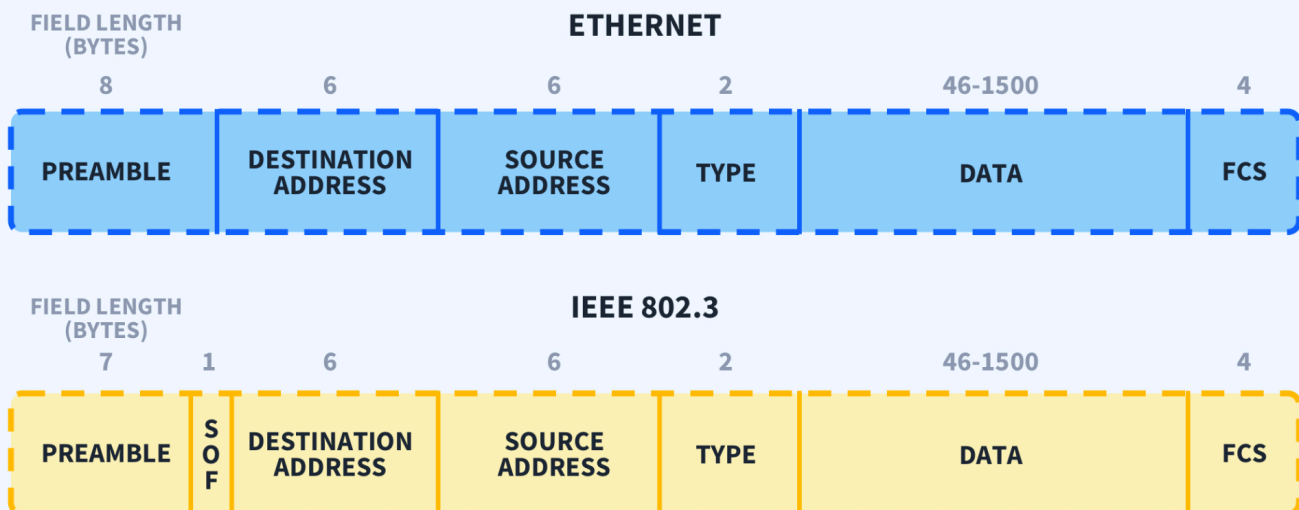


Ethernet Frame = The "Envelope" for Your Data

- It's the **format used to package and send data** over an Ethernet network.
- Contains important fields (sections) like:
 1. Preamble & Start Frame Delimiter (SFD) → Signals the start of the frame
 2. Destination MAC Address → Who the frame is going to
 3. Source MAC Address → Who sent the frame
 4. Type/Length → What kind of data is inside
 5. Data (Payload) → The actual message or content
 6. Frame Check Sequence (FCS) → Checks for errors in the frame



Each Ethernet device (like a laptop or switch) has a unique hardware address called a MAC address.





3. The Access Layer & Ethernet Switches



What is the Access Layer?

- The **first layer** where devices like **computers, phones, and printers connect to the network**, usually via Ethernet cables or Wi-Fi.
 - It's where your device **gets access to the local network (LAN)**.
-



4. How Do Ethernet Switches Work?



What is a Switch?

- A **network device** that **connects multiple devices** in a **local network (like a LAN)**.
- It **receives incoming data frames** and **forwards them to the correct device**.



How Does It Know Where to Send Data?

- It uses a special table called the **MAC Address Table**.
- It looks at the **Destination MAC Address** in the Ethernet frame to decide which port to send the data to.
- **Answer:** It adds devices' **MAC addresses** to its **MAC address table**, using the **Source MAC Address** field from incoming frames.

✓ Quick Summary Table: Key Concepts

Concept	What It Means	Why It's Important
Encapsulation	Wrapping data in a frame with address and control info	Ensures data is delivered properly over the network
Ethernet Frame	The "envelope" that carries data in an Ethernet network	Contains MAC addresses, data, and error-checking info
MAC Address	Unique hardware address of a network device (like a name tag)	Helps switches send data to the correct device
Switch	A device that connects multiple devices in a network & forwards data intelligently	Sends data only where it needs to go (more efficient than hubs)
Access Layer	The part of the network where devices connect (like your laptop to a switch)	Entry point for devices into the local network
Preamble & SFD	Signals the start of an Ethernet frame	Helps devices sync and detect the beginning of data
FCS (Frame Check Sequence)	Detects errors in the transmitted frame	Ensures data wasn't corrupted during transfer

Module(8)

1. What is an IPv4 Address? (Your Device's Online Address)

What Is It?

- An **IPv4 address** is a **unique number** assigned to every device (like a computer, phone, or printer) that connects to a **network** (like the internet or a local Wi-Fi).
- It works like your **home address** — it tells other devices **where to send data** (like emails, videos, or websites).

Why Do You Need One?

- To **send or receive data** over the internet or a local network (LAN), your device **must have an IPv4 address**.
- Every packet of data sent over the internet has:
 - A source IPv4 address (where it's coming from)
 - A destination IPv4 address (where it's going)

 **Without an IPv4 address, your device can't talk to other devices online or even on your local network.**



2. What Does an IPv4 Address Look Like?



IPv4 Address Format:

- An IPv4 address is made up of **32 binary bits**, but we write it in a much easier format called **dotted-decimal notation**.
- It looks like **four numbers** (each between 0–255), separated by **dots**.
- ♦ **Example:** 192.168.1.1 or 8.8.8.8 (Google’s public DNS server)

⚠ **Note:** You don’t need to know how to convert binary to decimal for now — just know that the friendly format you see is easier to read than 32 ones and zeros!



3. The Structure of an IPv4 Address



It Has Two Main Parts:

1. **Network Portion** → Identifies **which network** the device is on
2. **Host Portion** → Identifies the **specific device (host)** on that network



Think of it like a phone number:

- **Country code + Area code** = Network (where the phone is located)
 - **Local number** = Host (the specific person/device you’re calling)
-



Example:

Let’s say a device has this IPv4 address: 192.168.5.11

And the subnet mask is: 255.255.255.0

- The **first three numbers (192.168.5)** → **Network portion** (tells us which network the device is part of)
- The **last number (11)** → **Host portion** (identifies this specific device on that network)



All devices on the same network (with the same first 3 numbers) can talk to each other directly.



But to talk to devices on a different network, they need help from a router.

4. What is a Subnet Mask? (It Helps Identify the Network!)

- A subnet mask (like 255.255.255.0) is used along with the IPv4 address to figure out:
 - Which part is the network
 - Which part is the host

It's like a filter that separates the "street name" (network) from the "house number" (host).

5. Multiple Logical Networks on One Physical Network

- You can have **many smaller networks (logical networks)** on one **physical network** (like one office or home Wi-Fi).
- Devices with the **same network portion** in their IPv4 address can **talk to each other directly**.
- Devices with **different network portions must go through a router** to communicate.

 Real-life analogy:

Just like in a telephone system:

- The **country code + area code** = the **network**
- The **local phone number** = the **host (specific person/device)**

✓ Quick Summary: Key Points to Remember

Topic	What You Should Know
IPv4 Address	A unique 32-bit address that identifies a device on a network (like your home address online)
Format	Written as 4 numbers (0–255) separated by dots: e.g., 192.168.1.1 (called dotted-decimal notation)
Purpose	Used to send and receive data over the internet or local networks
Structure	Made of two parts: Network (where the device is located) and Host (the specific device)
Subnet Mask	Helps identify which part of the address is the network and which is the device
Routing	Routers use the network part of the address to send data to the correct network
Communication	Devices on the same network can talk directly. Devices on different networks need a router.

100%

IP address
Classes

(Class A)

0.1.8.1. IP

Subnet Mask	Network	Host
255	0	0 10

مکانی که در آن قرار می‌گیرد

* No of Net < 255

(Class B)

Subnet Mask	Network	Host
255	255	0 10

(Class C)

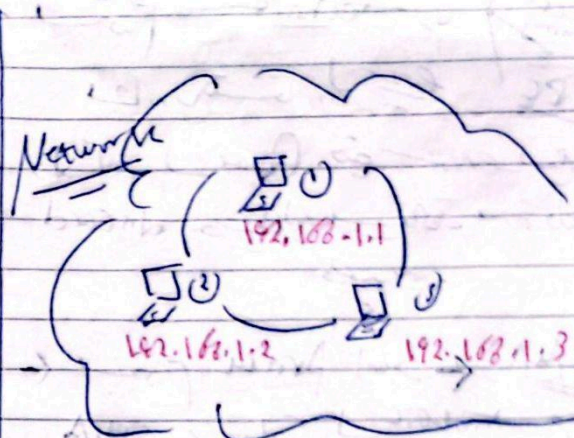
Network	Host
255 255 255	0

Many Networks

low No of Hosts

(Class D, E)

این کلاس‌ها برای



Here IP, Host & Network Class

* 192.168.1.0 → Network address

* 192.168.1.255 → broadcast address

این آدرس‌ها برای شبکه و پخش هستند

Types of IP address

Public

Private

این IP ها در اینترنت
و به ISP (We) می‌رسد

این IP ها در شبکه داخلی
و به روتر می‌رسد

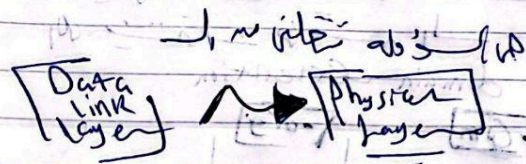
Lee 9

Mac add to Nic

* Each device has Hardware Component $\rightarrow$ Nic

* Nic (Network Interface Card)

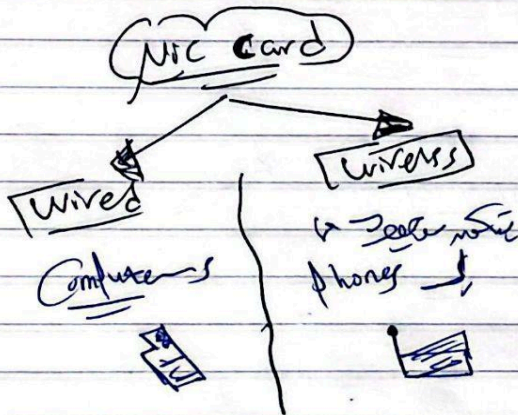
جهاز يربط الجهاز بالاسلاك



(0,1) is no data, device is in one or more medium signal

* Copper Cable $\rightarrow$ Electrical signal

* Opti Fiber $\rightarrow$ Light signal



جهاز يربط الجهاز بالاسلاك
 * Link/layer is device, wire,
 (Mac address), wire

* Mac address
 (Physical Address) *

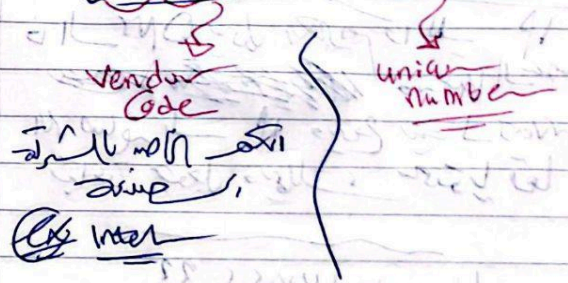
02-60-8D-39-1F-2A

$\rightarrow$ Octet (8 bit)

$\rightarrow$ hexa decimal

$\rightarrow$ length = 48 bit (6×8)

00-AA-02-BA-8F-5B



How I know mac address

Press win+R $\rightarrow$ write CMD
 write ipconfig

$\rightarrow$ ipconfig
 $\rightarrow$ ipconfig

Module(9)

Part 1: Types of IPv4 Addresses (Unicast, Broadcast, Multicast)

1. Unicast (One-to-One Communication)

- **Definition:** Sending data from **one device to exactly one other device**.
 - **Example:** Email sent from you to a friend.
 - IP Address Type:
 - Source IP: Always a unicast address
 - Destination IP: Also a unicast address
 - Range:
 - From 1.1.1.1 to 223.255.255.255 (but some are reserved for special use)
 - Default Type:
 - Most device-to-device communication is unicast, unless stated otherwise.
-

2. Broadcast (One-to-All Communication)

- **Definition:** Sending data from **one device to ALL devices on the same network**.
- **Example:** Announcing something to everyone in a room.
- IP Address Type:
 - Destination IP:
 - Limited Broadcast: 255.255.255.255 (sent to all devices on the local network)
 - Directed Broadcast: e.g., 172.16.4.255 (sent to all devices on a specific subnet like 172.16.4.0/24)
- How It Works:
 - All devices on the same broadcast domain (usually a local network) process the message.
- Important Notes:
 - Broadcasts are NOT used in IPv6.
 - Too many broadcasts can slow down the network.
 - Routers do NOT forward broadcasts → this helps isolate broadcast traffic.

✓ 3. Multicast (One-to-Selected-Group Communication)

- **Definition:** Sending data from **one device to a selected group of devices that “join” or subscribe to that group.**
 - **Example:** Sending a video stream only to users who want to watch it.
 - IP Address Range for Multicast:
 - 224.0.0.0 to 239.255.255.255
 - How It Works:
 - Devices interested in certain data join a multicast group.
 - The sender sends one copy of the data, and only members of the group receive it.
 - Used By:
 - Routing protocols like OSPF (uses multicast address 224.0.0.5)
 - Benefits:
 - Reduces unnecessary network traffic compared to broadcasting.
-

📖 Part 2: Types of IPv4 Address Categories

✓ 1. Public vs. Private IPv4 Addresses

Type	Definition	Can it go on the Internet?	Example Ranges
Private	Used inside organizations (homes, companies, schools)	✗ No (not routable on the Internet)	- 10.0.0.0 to 10.255.255.255 - 172.16.0.0 to 172.31.255.255 - 192.168.0.0 to 192.168.255.255
Public	Used for Internet communication	✓ Yes (routable on the Internet)	e.g., 8.8.8.8 (Google DNS), 192.0.3.15

- NAT (Network Address Translation):
 - Translates private IP → public IP when accessing the Internet.
 - Done by your router.

✓ 2. Special IPv4 Addresses

Type	Purpose	Example
Loopback	Used by a device to send data to itself (for testing)	127.0.0.1 (most common)
Link-Local (APIPA)	Automatically assigned when no DHCP server is found	169.254.0.0 to 169.254.255.254
Network & Broadcast Addresses	Cannot be assigned to devices	First & last IP in a subnet

✓ 3. Legacy: Classful Addressing (Old Method - Deprecated)

- IPv4 used to be divided into classes (A, B, C):
 - Class A: Big networks (1.0.0.0 to 126.0.0.0), supports millions of hosts
 - Class B: Medium networks (128.0.0.0 to 191.255.0.0), supports 65,000+ hosts
 - Class C: Small networks (192.0.0.0 to 223.255.255.0), supports 254 hosts
 - Class D: Multicast (224.0.0.0 to 239.255.255.255)
 - Class E: Experimental (240.0.0.0 to 255.255.255.255)

🔒 Classful addressing is no longer used.

✓ Today, we use Classless addressing (more efficient IP use).

✓ 4. Who Manages IP Addresses?

- **IANA** (Internet Assigned Numbers Authority) manages all IP addresses.
- It gives blocks of IPs to Regional Internet Registries (RIRs):
 - ARIN – North America
 - RIPE NCC – Europe, Middle East, Central Asia
 - APNIC – Asia-Pacific
 - LACNIC – Latin America & Caribbean
 - AfriNIC – Africa
- **RIRs** then give IPs to **ISPs**, who assign them to **organizations/companies**.

Part 3: Network Segmentation (Subnetting & Broadcast Domains)

1. What is a Broadcast Domain?

- A **group of devices** that can **receive each other's broadcast messages**.
 - **Switches** forward broadcasts to all connected devices.
 - **Routers DO NOT forward broadcasts** → they **separate broadcast domains**.
-

2. Why Segment a Network? (Why Subnet?)

- **Too many devices in one network = too many broadcasts = slow performance**
 - Benefits of Subnetting (Network Segmentation):
 1. Reduces broadcast traffic
 2. Improves network speed & performance
 3. Increases security (control which subnets can talk to each other)
 4. Better organization (group by location, function, device type)
-

3. How is Subnetting Done?

- **Subnets** are smaller networks created from a larger one.
- Achieved by using **some host bits as network bits** (longer subnet mask).
- Example:
 - Large Network: 172.16.0.0/16 (65,000+ hosts)
 - Split into:
 - Subnet 1: 172.16.0.0/24 (200 users)
 - Subnet 2: 172.16.1.0/24 (200 users)

 Longer prefix (e.g., /24 instead of /16) = Smaller network = Fewer hosts per subnet

✓ 4. Common Ways to Divide Subnets

Method	Description	Example
By Location	Different floors or buildings get different subnets	Floor 1: 10.0.1.0/24, Floor 2: 10.0.2.0/24
By Function/Group	Group by department (HR, IT, Finance)	HR: 10.0.1.0/24, IT: 10.0.2.0/24
By Device Type	Separate printers, servers, users	Servers: 10.0.1.0/24, Printers: 10.0.2.0/24

✓ Check Your Understanding – Quick Recap Questions

? About Broadcasts:

- Which device does NOT forward broadcasts by default?
👉 Router
- What problems are caused by too many broadcasts?
👉 Slow network & device performance

? About IP Types:

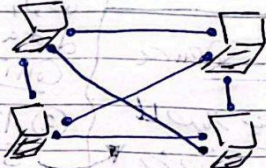
- Which IP types can go on the Internet?
👉 Public IPs
- Which IP types are used inside organizations?
👉 Private IPs
- Which organization assigns IPs to ISPs?
👉 RIRs (via IANA)

Summary in One Slide (Easy to Remember)

Concept	Key Points
Unicast	One-to-One, used always by default, source & dest = unicast
Broadcast	One-to-All, dest = 255.255.255.255 or directed, routers block it
Multicast	One-to-Group, efficient, used by apps like OSPF
Private IP	For internal use only, not on Internet (e.g., 192.168.x.x)
Public IP	For Internet, globally unique
NAT	Converts private ↔ public IP for Internet access
Subnetting	Splits big networks into smaller ones to reduce broadcasts
Broadcast Domain	Area where devices receive each other's broadcasts
Routers	Separate broadcast domains, improve performance

(3) Mesh Topology

Every Cable → (n-1) × (n-2) / 2



Adv

① No Fault Tolerance
if one cable is damaged, the whole network is affected.

③ More Secure
Cable not required for every node to every node.

Disadv

① Very High Cost
→ A lot of Cables
→ Network Interfaces

② difficult to install and wire

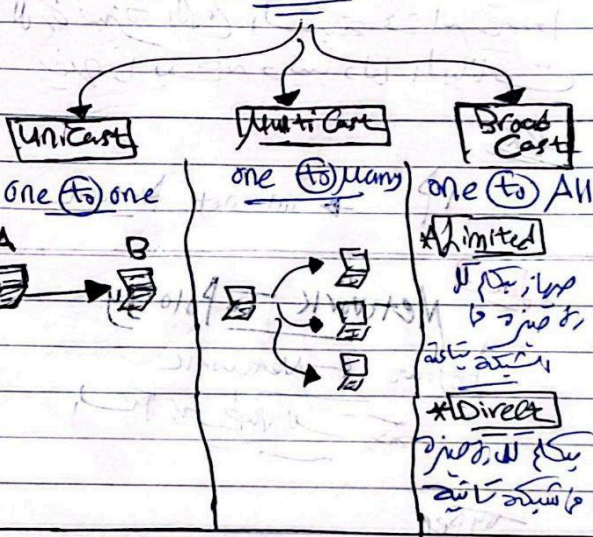
③ difficult to manage and troubleshoot

Handwritten notes in Urdu: "بہت زیادہ مہنگا", "بہت زیادہ کابلیں", "بہت زیادہ مینیجنگ", "بہت زیادہ ٹراؤبلسھوٹنگ"

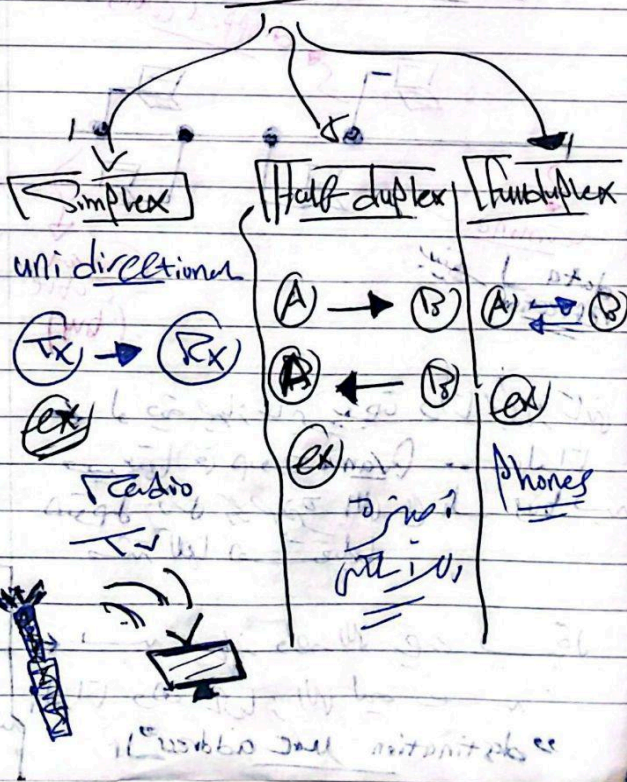
lec 4

Types of Casting

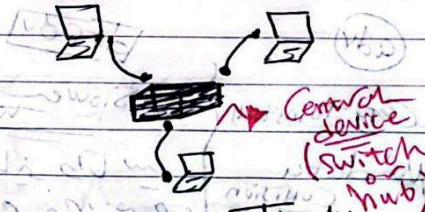
Casting → transmitting data over the network



Transmission Modes



(4) Star Topology (Most Common)



Adv

① Inexpensive
Fewer cables than mesh topology.

② Scalability
Easy to add or remove nodes.

Disadv

① Single Point of Failure
If central device fails, the whole network is affected.

Handwritten notes in Urdu: "اگر مرکزی ڈیوائس خراب ہو جائے تو پوری نیٹ ورک متاثر ہو جائے گی", "ایک ہی جگہ پر خرابی ہو سکتی ہے"

Module(10)

Introduction: Why IPv6?

You already know about IPv4 — the current system that gives every device on the internet a unique address.

But here's the problem: IPv4 is running out of addresses!

👉 That's why the world is moving to **IPv6**, the next-generation Internet Protocol.

Part 1: Why We Need IPv6 (IPv4 Issues)

! The Main Problem: IPv4 Address Depletion

- IPv4 uses **32-bit addresses** → ~4.3 billion possible addresses
- But the world has **billions of devices** now: phones, laptops, smart TVs, cars, sensors, etc.
- Four out of five Regional Internet Registries (RIRs) have already run out of IPv4 addresses:
 - ARIN (North America) → July 2015
 - RIPE NCC (Europe) → September 2012
 - APNIC (Asia-Pacific) → June 2014
 - LACNIC (Latin America) → April 2011
 - AfriNIC (Africa) → Projected 2020

Why the Demand?

- Growth in **mobile devices** (over 90% of mobile traffic in the US is over IPv6!)
- Expansion of **Internet of Things (IoT)** → smart homes, cars, wearables, industrial sensors
- NAT (Network Address Translation) is used to share IPv4 addresses, but:
 - It causes latency
 - Breaks peer-to-peer communication
 - Makes some apps not work well

What is IPv6?

- **IPv6 is the successor to IPv4**
- Uses **128-bit addresses** → **340 undecillion (340×10^{36}) possible addresses** → **practically unlimited**
- Developed by **IETF** to **fix IPv4 limitations** and add new features

IPv6 Improvements Over IPv4:

Feature	IPv4	IPv6
Address Length	32-bit (~4.3B addresses)	128-bit (340 undecillion addresses)
Addressing Issue	Running out	More than enough
Auto-Configuration	Limited	Built-in (via ICMPv6)
Broadcast	Yes	 No (uses multicast instead)
Security	Optional (IPSec)	Built-in support
Header Simplicity	Complex	Simplified, faster processing

Part 2: How IPv4 and IPv6 Will Coexist (Transition Mechanisms)

Since the change to IPv6 won't happen overnight, **IPv4 and IPv6 will coexist for years.**

The **IETF** has created **3 main transition methods**:

1 Dual Stack (Best Option)

- **Both IPv4 and IPv6 run at the same time** on the same device/router/network.
- Devices can communicate using **either IPv4 or IPv6 natively.**
- Preferred method for **direct IPv6 communication.**

 How it works:

Your device has two protocol stacks (IPv4 & IPv6), and chooses which one to use based on the destination.

2 Tunneling

- Used to send **IPv6 packets over an IPv4 network.**
- The **IPv6 packet is wrapped (encapsulated) inside an IPv4 packet.**
- Like putting a letter (IPv6) inside an envelope (IPv4) to send it through the mail.

 Use case:

Connecting two IPv6-only networks over an IPv4-only internet.

3 Translation (NAT64)

- Allows **IPv6 devices to talk to IPv4 devices** and vice versa.
- Similar to how **NAT works for IPv4**, but for **IPv6 ↔ IPv4 translation.**
- Used when one side is IPv6 and the other is IPv4.

 How it works:

An IPv6 packet is translated into IPv4, sent across the network, then translated back.

 Note:

Tunneling and Translation are temporary solutions. The final goal is native IPv6 communication (end-to-end).

Part 3: IPv6 Addressing Basics

1. Hexadecimal Number System (Base 16)

- IPv6 uses **hexadecimal** (0–9 and A–F)
- Each digit = **4 bits**
- 128-bit address = **32 hexadecimal digits**, written in **8 groups of 4 hex digits (called hextets)**

 **Digits used:** 0 1 2 3 4 5 6 7 8 9 A B C D E F

2. IPv6 Address Format

- Written as: **x:x:x:x:x:x:x** (8 hextets)
- Each “x” = 16 bits (4 hex digits)
- **Hextet** = a group of 4 hexadecimal digits (16 bits)
- **Not case-sensitive** (A = a)

♦ **Example (Preferred Format):**

2001:0db8:0000:0000:0000:ff00:0042:8329

3. Two Rules to Shorten IPv6 Addresses

✓ Rule 1: Omit Leading Zeros

- You can remove zeros at the **start of each hextet**

♦ Before:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

♦ After (Omit leading zeros):

```
2001:db8:0:0:0:ff00:42:8329
```

✓ Rule 2: Double Colon (::) – Replace Consecutive Zero Hextets

- You can replace **one or more consecutive groups of zeros** with ::
- But **only once per address!**

♦ Before:

```
2001:0db8:0000:0000:0000:0000:0000:0001
```

♦ After (Compressed with ::):

```
2001:0db8::1
```

 **Important:** Don't use :: more than once, or it becomes unclear how many zeros were removed.

Part 4: Check Your Understanding – Quick Review Questions

IPv4 Issues

1. What is the biggest reason for moving to IPv6?
 Depletion of IPv4 addresses
2. True or False: 4 out of 5 RIRs have run out of IPv4 addresses?
 True
3. Which method provides native IPv6 connectivity?
 Dual Stack

Summary: IPv6 in a Nutshell

Topic	Details
Why IPv6?	IPv4 is out of addresses; IPv6 has 340 undecillion
Key Feature	128-bit addresses, better performance, auto-configuration, no broadcasts
Coexistence Methods	Dual Stack (best), Tunneling, Translation (NAT64)
IPv6 Address Length	128 bits (vs 32 in IPv4)
Address Format	8 hextets (4 hex digits each), can be shortened
Shortening Rules	Omit leading zeros, use :: for consecutive zeros (once only)

Module(11)

Introduction: Why Learn About DHCP?

Imagine this:

sherif tries to log in to a new hospital laptop to access patient files, but it can't connect to the network.

After checking, the IT technician, Madhav, finds out that someone manually set a wrong IP address on the laptop.

🤔 How can we avoid these manual mistakes?

👉 The answer is: DHCP (Dynamic Host Configuration Protocol) — a system that automatically gives devices the correct network settings like IP addresses.

Part 1: Static vs. Dynamic IP Addressing

✓ 1. Static IP Address Assignment (Manual)

- **Definition:**
A network administrator manually enters all the network settings on a device (like IP address, subnet mask, default gateway).
- **What's Configured Manually?**
 - **IP Address** – Identifies the device on the network
 - **Subnet Mask** – Defines the network range
 - **Default Gateway** – The router that connects to the internet or other networks
- **Advantages:**
 - Good for **servers, printers, or network devices** that need a **fixed address** (so users can always find them)
 - More **control** over which device has which address
- **Disadvantages:**
 - **Time-consuming** to configure each device manually
 - Higher chance of **human error** (wrong IP, typos, etc.)
 - Must keep a **list of who has which IP**
 - IP addresses are **permanent** and not reused easily

🔧 Best for:

Devices that must always have the same address (e.g., file servers, network printers)

✓ 2. Dynamic IP Address Assignment (Automatic)

- **Definition:**
IP settings are automatically given to devices by a DHCP server when they connect to the network.
- **What's Configured Automatically?**
 - **IP Address**
 - **Subnet Mask**
 - **Default Gateway**
 - **DNS Servers** (not mentioned here but also assigned by DHCP)
- **How it Works:**
 - Devices (like laptops, phones) have a **DHCP Client** built in.
 - When you connect to Wi-Fi or a network, your device **asks a DHCP server** for an IP address.
 - The **DHCP server assigns** an available IP from a **pool** of addresses.
- **Advantages:**
 - **No manual setup** needed for each new device
 - **Reduces human errors**
 - **Saves time** for IT staff
 - IP addresses are **leased temporarily** and can be **reused** when not in use (great for mobile users)
- **Disadvantages:**
 - Not ideal for devices that **must always have the same IP** (though DHCP can be configured for that too)

🔧 Best for:

- **Large networks** (offices, airports, cafes)
- **Mobile users** (laptops, phones that connect/disconnect often)
- **Home networks, Wi-Fi hotspots, corporate offices**

3. What is a DHCP Server?

- A **DHCP Server** is a device or software that **automatically gives IP configurations** to devices (clients) on the network.
- **Where can DHCP servers be located?**
 - **In medium/large offices:** Often a **dedicated server or PC**
 - **At home or in small businesses:** Usually your **wireless router** (it gets a public IP from your ISP, then gives private IPs to your devices)
 - **At ISPs (Internet Providers):** They may assign IPs directly or through a DHCP server
 - **Other devices:** Routers or switches can also act as DHCP servers (less common)

Real-life Example:

- When you go to a **coffee shop or airport**, your laptop automatically gets internet access because the **local Wi-Fi router (DHCP server)** gives your device an IP address.

Comparison Table: Static vs. Dynamic IP

Feature	Static IP	Dynamic IP (DHCP)
Assigned By	Network admin (manual)	DHCP server (automatic)
Setup Time	Slow (manual per device)	Fast (auto)
Error Risk	High (typos, wrong info)	Low (auto-configured)
Reuse of IPs	Rare (fixed addresses)	Yes (IPs are leased & recycled)
Best For	Servers, printers, network devices	Laptops, mobile devices, general users
Management	Needs IP tracking & maintenance	Hands-off, easy to scale

✓ 4. Check Your Understanding – Quick Review Questions

? Question 1:

Who is responsible for statically assigning IP addressing information?

✓ Correct Answer: The network administrator

? Question 2:

What protocol is responsible for automatically assigning IP addressing information?

✓ Correct Answer: DHCP

🎯 Summary – DHCP in Simple Words

Concept	What It Means
Static IP	You (or an admin) manually set the IP. Good for things that must always stay the same (like a server).
Dynamic IP	The DHCP server gives you an IP automatically when you connect. Great for users and devices that come and go .
DHCP	A protocol/service that automatically hands out IP addresses and network settings to devices.
DHCP Server	The device (like a router or server) that assigns IPs automatically . Found in homes, offices, Wi-Fi hotspots.
Why DHCP is Awesome	Saves time, avoids mistakes, works automatically, and is perfect for today's mobile and growing networks.

DHCP

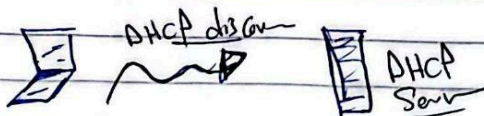
Router Service ←
 ←
 Automatic IP address

IP ←
 DHCP Server

4 Steps → DoRA process

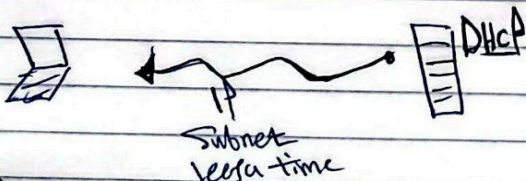
① Discover

IP ← DHCP Server



② Offer

← DHCP Server
 ←



③ Request

←
 ←

④ Acknowledgment

←
 (DHCP)

Note

← DHCP Server
 ←

← IP

lease time →

← IP

←

←
 10.0.0.5 ← 10.0.0.1
 ←

← lease time

←
 Complete

← lease time

←
 ←

←
 ←

Module(12)

Introduction: Why Learn About Gateways & NAT?

In this module, sherif learns how **devices in one network** (like a hospital department) **communicate with the outside world** (like the Internet or other departments).

She discovers that **gateways and a technology called NAT (Network Address Translation)** make this possible — allowing seamless email, web browsing, and online communication **even when using private (non-public) IP addresses**.

Part 1: Network Boundaries & Routers as Gateways

1. What is a Gateway?

- A **gateway** is a **router interface** that connects your **local network to other networks** (like the Internet).
 - It acts as the **“exit door”** for your devices to reach outside networks.
-

2. Routers as Gateways

- Every **host (computer, phone, printer)** on your **local network** uses a **default gateway** to send data **outside the local network**.
- The **default gateway is the IP address of the router’s interface** that’s connected to your local network.
- This address can be:
 - **Set manually (statically)** by an admin
 - **Assigned automatically (dynamically)** via **DHCP**

Example:

- Local computer IP: 192.168.1.101
- Subnet Mask: 255.255.255.0
- **Default Gateway (Router):** 192.168.1.1

➡ All devices on the 192.168.1.0 network send **internet-bound traffic to 192.168.1.1**

✓ 3. Routers as Network Boundaries

- The **router separates two networks**:
 - **Inside (Internal/Private) Network**: Your home, office, or hospital LAN (uses **private IP addresses** like 192.168.x.x)
 - **Outside (External/Public) Network**: The Internet (uses **public IP addresses**)

🔒 Private IPs (inside) are not routable on the Internet.
So, the router (gateway) acts as the bridge between them.

✓ 4. How Does It Work in Real Life?

- Your **wireless router** (at home or office):
 - Gets a **public IP from your Internet Service Provider (ISP)**
 - Gives **private IPs to your devices** (like 192.168.1.101) using **DHCP**
 - Acts as the **DHCP server** (gives IPs, subnet masks, and the **default gateway**)
 - Acts as the **DHCP client** (gets a public IP from the ISP)

Example Topology:

- **Internet (Cloud) ↔ ISP Router ↔ Wireless Router (Your Gateway)**
 - **Internal Network (LAN)**: Devices with private IPs (192.168.1.x)
 - **External Network (WAN/Internet)**: Public IP on the router's other interface (e.g., 192.150.45.3)

✓ Check Your Understanding – Network Boundaries

? Question 1: Same Network (2 Hosts)

Which statements are TRUE for two devices on the **same local network**?

✓ Correct Answers:

- Both hosts will have the **same default gateway address**
- Both hosts will have **different IP addresses**
- Both hosts will have **different MAC addresses**

✗ Incorrect:

- Same IP, same MAC, or different gateways (not typical on same network)
-

? Question 2: Different Networks (2 Hosts)

Which statements are TRUE for two devices on **different networks**?

✓ Correct Answers:

- Both hosts will have **different IP addresses**
- Both hosts will have **different default gateway addresses**
- Both hosts will have **different MAC addresses**

Part 2: Network Address Translation (NAT)

What is NAT (Network Address Translation)?

- NAT is a **technology used by routers** to allow **multiple devices on a private (local) network** to share a **single public IP address** when accessing the Internet.

Why is NAT needed?

- **Private IPs (like 192.168.x.x) cannot be used on the Internet.**
- But your **ISP gives you only ONE public IP** (for your router).
- **NAT solves this by translating:**
 - **Private IPs → Public IP** when going out to the Internet
 - **Public IP → Private IP** when responses come back

Result:

All your devices (phones, laptops, printers) can access the Internet using just one public IP assigned to your router.

How NAT Works (Simplified):

1. Your **device (192.168.1.101)** wants to visit a website.
2. It sends the request to the **router (default gateway 192.168.1.1)**.
3. The **router replaces your private IP with its public IP (e.g., 192.150.45.3)**.
4. The **response comes back to the public IP**, and the router **sends it back to your private IP**.

 This all happens automatically and **quickly** thanks to NAT.

Summary: Key Concepts

Concept	What It Means
Gateway (Default Gateway)	The router's IP address on your local network that lets devices access other networks (like the Internet)
Local (Internal) Network	Your home, office, or hospital network — uses private IPs (e.g., 192.168.x.x)
External (Public) Network	The Internet — uses public IPs assigned by your ISP
Router as Boundary	The router separates your internal private network from the external public Internet
DHCP	Gives your devices IPs, subnet masks, and the default gateway automatically
NAT (Network Address Translation)	Allows many devices to share one public IP to access the Internet by translating private ↔ public IPs

Quick Review Checklist

- ✓ A gateway (router) is needed to connect your local network to the Internet
- ✓ The default gateway IP is how devices send traffic outside their network
- ✓ Routers separate internal (private) and external (public) networks
- ✓ Private IPs (like 192.168.x.x) can't be used on the Internet — NAT fixes that
- ✓ NAT lets many devices share one public IP for Internet access
- ✓ DHCP automatically gives your device the right network settings, including the default gateway

Module(13)

Introduction: Why This Matters

sherif notices something interesting:

- Her **phone gets a different IP address** when she connects to **different Wi-Fi networks** (like at home, the hospital, or a café).
- But her **MAC address stays the same no matter where she is**.

 **Why is that?** Because:

- **IP address** tells the network **where the device is located** (on which network).
- **MAC address** is **unique to the device itself** (like its fingerprint) and never changes.

She also wonders:

How does data find the right device on a network? That's where **MAC and IP addressing, ARP, and broadcast domains** come in!

Part 1: MAC Addresses vs. IP Addresses

1. IP Address (Internet Protocol)

- Tells the **network where to send the data** (like a **postal address**).
- Can **change** depending on the network you're connected to (thanks to **DHCP**).
- Used to route data **between networks** (like from your home to the internet).

♦ **Example:** - At home: 192.168.1.10

- At hospital: 10.0.0.15
- On mobile data: 172.56.30.42

✓ 2. MAC Address (Media Access Control)

- A **unique hardware address** assigned to your **network card (NIC)**.
 - Never changes, no matter which Wi-Fi or network you connect to.
 - Used to **deliver data to the correct device on the same local network**.
- ♦ **Example:** - Your phone's MAC might be: aa-aa-aa-aa-aa-aa
- It's **burned into the device** and is unique worldwide.
-

↻ How They Work Together

Step	What Happens	Address Used
1	You send data to another device (by IP)	IP Address tells where the device is
2	Data reaches the local network	MAC Address tells which exact device gets it
3	The router uses the IP to route data across networks	MAC is used inside each local network

Part 2: How Devices Find Each Other's MAC Address (ARP & ND)

Problem:

You know the **IP address** of the device you want to talk to (like 192.168.1.10), but to send data on a local network, you also need its **MAC address**.

Solution 1: ARP (Address Resolution Protocol) – for IPv4

- ARP is used to **find the MAC address of a device on the same local network when you only know its IP**.

How ARP Works (3 Steps):

1. Your device sends a **broadcast message** (to all devices): “Who has IP 192.168.1.10? Tell me your MAC!”
2. The device with that IP replies: “I’m 192.168.1.10, and my MAC is aa-aa-aa-aa-aa-aa.”
3. Your device **remembers (caches)** this info in an **ARP Table** for next time.

 **Note:** ARP only works **within the same local network (broadcast domain)**.

Solution 2: NDP (Neighbor Discovery Protocol) – for IPv6

- IPv6 uses **NDP (ICMPv6 Neighbor Discovery)** instead of ARP to do the same job.

Part 3: Broadcast Domains & How They're Managed

1. What is a Broadcast Domain?

- A **broadcast domain** is a **local network** (usually one or more switches) where:
 - When a device sends a **broadcast message**, **all devices on that network receive it**.
 - Broadcasts are used for things like:
 - Finding the MAC address (via ARP)
 - DHCP requests
 - Some network announcements

 Too many broadcasts = slower network performance

2. How Do We Control Broadcasts?

- **Switches** forward broadcasts to **all connected devices** in the same network.
- **Routers do NOT forward broadcasts.**
- Therefore, **routers are used to split one big network into smaller broadcast domains.**

 **Result:** - Better performance

- Less unnecessary traffic
- More **organized and secure** networks

 **Example:**

- **3 broadcast domains:** Production, Sales, Marketing
 - 👉 Each has its own **switches and devices**, and they're connected via a **router**

Part 4: How Devices Communicate on the Same Network

Scenario:

- **PC1** (IP: 192.168.10.10) wants to send data to **PC2** (IP: 192.168.10.11)
- They're on the **same local network**

What's Inside the Network Frame?

Layer	Field	Value (Example)
Layer 2 (Ethernet Frame)	Destination MAC	55-55-55 (PC2's MAC)
	Source MAC	aa-aa-aa (PC1's MAC)
Layer 3 (IP Packet)	Destination IP	192.168.10.11 (PC2)
	Source IP	192.168.10.10 (PC1)

 How does PC1 know PC2's MAC? 🙋 It uses **ARP** to find it!

Check Your Understanding – Quick Review Questions

Question 1:

What destination MAC address is used when sending to a device on the same local network?  Correct Answer: The MAC address of the destination device

Question 2:

What destination MAC address is used when sending to a device on a remote network?  Correct Answer: The MAC address of the local router interface (default gateway)

Question 3:

Which protocols are used to find a device's MAC address from its IP?  Correct Answers: - ARP (for IPv4)

- ND (Neighbor Discovery) (for IPv6)

Summary Table: Key Concepts

Concept	What It Means
IP Address	Logical address that tells where the device is located (can change per network)
MAC Address	Physical/hardware address unique to each device (never changes)
ARP (Address Resolution Protocol)	Used to find a device's MAC address from its IP (on the same local IPv4 network)
ND (Neighbor Discovery)	IPv6 version of ARP
Broadcast Domain	A local network where broadcasts reach all devices (managed by switches)
Router	Separates broadcast domains, routes traffic between networks, and forwards data based on IP
Default Gateway	The router's IP on your local network — used to send data outside your local network

Quick Tips to Remember

- ✓ **IP = Location** (changes per Wi-Fi/network)
- ✓ **MAC = Identity** (unique to your device, never changes)
- ✓ **ARP = Finds MAC from IP (on same network)** ✓ **Routers block broadcasts & connect networks** ✓ **Switches forward broadcasts within the same network**

Module(14)



Introduction: Why This Topic Matters

Sherif is driving home and uses GPS to avoid a traffic jam by taking a faster, less congested route.

She then wonders:

"Can computer networks get congested too? And do they find better paths, like a GPS does?"

✓ Yes, they can — and that's where routing comes in.

Just like a GPS calculates the best road, routers in a network find the best path for your data to travel — especially when going outside your local network.



Part 1: The Need for Routing



What is Routing?

- **Routing** is the process of finding the **best path** for data (called packets) to travel from a **source device to a destination device**, especially when they are on **different networks**.
-



1. What Does a Router Do?

- A **router** is a device that **connects multiple networks** (like your home Wi-Fi to the Internet).
- It works at **Layer 3 (Network Layer)** of the networking model and uses **IP addresses** (not MAC addresses like switches).
- Routers help **send data between networks**, such as:
 - From your laptop to Google (which is on the Internet)
 - From one office building to another

How?

- The router **looks at the destination IP address** in the packet.
 - It then decides **which network path** (interface or next router) is the best to send the data.
-

2. When Do You Need a Router?

- Whenever a device wants to send data to another device on a **different network** (i.e., their IP addresses are in **different network ranges**).
 - Example:
 - Device on 1.1.1.0 network → sending to 5.5.5.0 network → **must go through a router**
-

Part 2: The Routing Table

What is a Routing Table?

- A **routing table** is the **router's brain** — it's a list that tells the router:
 - **Which networks it knows about**
 - **The best way (path) to reach each of those networks**
-

What if No Route is Found?

- The router will **drop the packet (delete it)** if it doesn't know where to send it.
- That's why having a **default route** is important — it helps avoid lost data.

Part 3: The Default Gateway

What is a Default Gateway?

- The **default gateway** is the **IP address of the router** on your **local network**.
 - It's the path your **computer or phone** uses to **send data to other networks** (like the Internet).
-

How Does It Work?

1. Your device wants to send data to a remote network.
 2. It sends the data **to the router (default gateway)**.
 3. The packet is put into an **Ethernet frame**, but this time, the **destination MAC address is the router's MAC**, not the final device.
 4. The **router gets the packet**, looks at the **destination IP**, and forwards it to the right place.
-

How Does Your Device Know the Router's MAC Address?

- Your device uses **ARP (Address Resolution Protocol)** to find the **MAC address of the router**, using its **IP (default gateway)**.
-

Important:

- If your device **doesn't have a default gateway set**, or it's wrong, **you can't access anything outside your local network** (like the web, email, or cloud apps).

Part 4: Local Area Networks (LANs)

What is a LAN?

- A **LAN (Local Area Network)** is a network that covers a **small area**, like:
 - A **home**
 - An **office**
 - A **school or building**
- All devices on a LAN are usually under **one administration** (one company or person controls it).
- LANs use technologies like **Ethernet (wired)** or **Wi-Fi (wireless)** and support **fast data speeds**.

 **Intranet:** A private LAN for use only by an organization's members.

Single vs. Multiple Network Segments in a LAN

♦ All Hosts on One Local Network (Single Segment)

Pros:

- Simple to set up
- Low cost
- Fast communication (direct device-to-device)
- Easy to find other devices (via ARP)
- All devices can communicate easily

Cons:

- All devices share **one broadcast domain** → can cause **traffic congestion**
- Harder to manage **security** or **quality of service (QoS)**

 Best for: **Small offices, homes, or simple networks**

◆ Hosts Split Across Multiple Networks (Remote Segments)

✓ Pros:

- Better **performance** (less broadcast traffic)
- Improved **security**
- Easier **organization & management**
- Broadcasts are contained

✗ Cons:

- Requires **routing (a router or Layer 3 switch)**
- Adds **complexity and cost**
- Slight delay (latency) when routing between segments

📌 Best for: **Larger networks, businesses, or organizations with many devices**

✓ Check Your Understanding – Key Concepts

Concept	What It Means
Routing	Finding the best path for data to travel between networks
Router	Device that connects and directs traffic between different networks using IP addresses
Routing Table	A list of known networks and how to reach them, used by routers
Default Route	A backup path used when a router doesn't know where to send data
Default Gateway	The router's IP on your local network — used to send data to other networks
LAN (Local Area Network)	A local, high-speed network under one administration, like at home or in an office
Broadcast Domain	A network area where all devices receive broadcast messages (can slow things down if too large)

Simple Summary in Everyday Terms

Topic	Simple Explanation
Why Routing?	Just like a GPS finds the fastest road, routers find the best path for your data to travel across networks.
What is a Router?	A device that connects different networks and decides where to send your data based on IP addresses.
Routing Table	The router's guidebook that lists all known networks and the best way to reach them.
Default Gateway	The router your device uses to send data to the Internet or other outside networks.
LAN	Your home or office network — all your devices connected locally, usually via Wi-Fi or cables.
Single vs. Multiple Segments	Keeping all devices on one network is simple but can get crowded; splitting them helps with speed and security.

Module(15)

Introduction: Why This Module Matters

sherif is in an important video conference call at work using her hardwired desktop computer, but she notices some dropped words or lag.

She wonders:

“Is this a network issue?”

Madhav from IT explains that this might be related to the type of transport protocol being used — specifically UDP, which is commonly used for video calls.

He also mentions TCP, which is used for more reliable communication.

🤔 What’s the difference?

Let’s break it down!

Part 1: TCP vs. UDP (Transport Layer Protocols)

Where Do They Fit?

- Both **TCP (Transmission Control Protocol)** and **UDP (User Datagram Protocol)** operate at the **Transport Layer (Layer 4)** of the **TCP/IP model**.
 - Their job is to **manage how data is sent between applications** running on different devices.
-

1. TCP (Transmission Control Protocol) – Reliable Delivery

- **"Do-it-right" protocol**
- **Ensures reliable, ordered delivery** of data
- **Keeps track of every data segment (packet)** sent
- If a packet is **lost or corrupted**, **TCP resends it**
- Used for applications where **accuracy is critical**, such as:
 - Web browsing (HTTP/HTTPS)
 - Email (SMTP)
 - File transfers (FTP)
 - Online banking

Features:

- **Acknowledgement** of received data
 - **Error checking**
 - **Sequencing** of packets
 - **Connection-oriented** (sets up a formal connection before sending data)
-

2. UDP (User Datagram Protocol) – Fast & Lightweight

- **"Best effort" protocol**
- **Does NOT guarantee delivery**, order, or error-checking
- **Faster** than TCP because it has **less overhead**
- Used for applications where **speed matters more than reliability**, such as:
 - **Video calls / conferencing**
 - **Voice over IP (VoIP)**
 - **Live streaming (like YouTube Live or Twitch)**
 - **Online gaming**

Why UDP?

Occasional loss (like a blurred second in a video call) is more acceptable than waiting for delays caused by resending lost data.

TCP or UDP?

Use Case	Best Protocol
Watching a live sports stream	UDP
Downloading a file	TCP
Sending an email	TCP
Voice chat (Zoom, Teams, Skype)	UDP
Browsing the web	TCP

Part 2: Port Numbers

What is a Port Number?

- A **port number** is used to **identify specific services or applications** on a device.
- It works like a "**door number**" that data is sent to on a computer or server.
- Every message sent using **TCP or UDP** includes:
 - A **source port** (where it came from)
 - A **destination port** (where it's going)

 Ports help the device know which app the data belongs to.

Common Well-Known Ports:

Application	Port Number	Protocol
HTTP (web pages)	80	TCP
HTTPS (secure web)	443	TCP
FTP (file transfer)	21	TCP
SMTP (email sending)	25	TCP
DNS (domain names)	53	UDP (sometimes TCP)
DHCP	67/68	UDP

Pro Tip:

You can visit the IANA website to see the full official list of port numbers and their assigned services.

✓ Some Apps Use Both TCP and UDP:

- **DNS** → Uses **UDP** for quick client requests, but **TCP** for server-to-server communication
 - **VoIP & Video** → Mostly **UDP** for speed, but may use **TCP** for control signals
-

📘 Part 3: Socket Pairs (Identifying Unique Connections)

✓ What is a Socket?

- A **socket** is the combination of:
 - An **IP address**
 - A **port number**

🔗 It uniquely identifies a specific conversation between a client and a server.

🧩 Socket Pair Example:

When your **PC (192.168.1.5)** connects to a **web server (192.168.1.7)**:

- **Client (your PC):** 192.168.1.5:1099 (source IP + random port)
- **Server (web service):** 192.168.1.7:80 (destination IP + port 80 for HTTP)

👉 Together, they form a socket pair:

192.168.1.5:1099 ↔ 192.168.1.7:80

This helps the server send the **correct response back to your browser** (not someone else's!).

Why Are Ports & Sockets Important?

- Allow **multiple apps/services** to run on the **same device** at the same time
 - Help the **transport layer** send data to the **correct application**
 - Enable **multiple connections** to the **same server service** (like many people visiting the same website)
-

Part 4: The netstat Command (Windows)

What is netstat?

- A **command-line tool** used to display:
 - **Active network connections**
 - **Protocols in use (TCP/UDP)**
 - **Local and remote IP addresses & ports**
 - **Connection status (e.g., ESTABLISHED, LISTENING)**
-

Example Output:

Bash

C:\> netstat

Active Connections

Proto	Local Address	Foreign Address	State
TCP	192.168.1.124:3126	192.168.0.2:netbios-ssn	ESTABLISHED
TCP	192.168.1.124:3158	207.138.126.152:http	ESTABLISHED
TCP	192.168.1.124:3166	www.cisco.com:http	ESTABLISHED

Why Use netstat?

- Helps detect **unwanted or suspicious connections**
 - Useful for **troubleshooting network issues**
 - Can confirm which **apps are using the network**
-

Tip:

Use netstat -n to show **numeric addresses & ports** (no DNS or service name lookups).

Check Your Understanding – Quick Review Questions

TCP and UDP

Q1: Which layer ensures packets are sent reliably and resends missing ones?

 Answer: Transport Layer

Q2: Does TCP keep track of sent segments?

 Answer: True (TCP tracks and resends if needed)

Port Numbers

Q1: What is the correct socket pair for a web request from 10.1.1.10 to server 10.1.1.254 (port 80)?

 Answer: 10.1.1.10:1099, 10.1.1.254:80

Q2: Which ports are used for FTP, HTTP, and TFTP?

 Answer: Well-known ports (typically below 1024)

Q3: Which command shows active connections, ports, and states on Windows?

 Answer: netstat

Summary Table: Everything You Need to Know

Topic	Key Points
TCP	Reliable, connection-oriented, ensures delivery, used for web, email, file transfers
UDP	Fast, connectionless, no delivery guarantee, used for video, VoIP, gaming
Port Numbers	Identify specific apps/services (e.g., port 80 = HTTP, port 53 = DNS)
Sockets	Pair of IP + Port that defines a unique connection (e.g., 192.168.1.5:1099 ↔ 192.168.1.7:80)
netstat	Command to view active network connections, protocols, ports, and statuses
Transport Layer (Layer 4)	Where TCP and UDP operate; manages end-to-end communication

Module(16)

Introduction: How Do You Actually Access Stuff on the Internet?

sherifi is at work and needs to access a patient file on the hospital's network.

She's done it many times before — but now she's curious:

"How does this actually work? How do I get files from the intranet or the internet? What's happening behind the scenes?"

✓ Great question — it's all thanks to *application layer services* and the way *clients and servers* communicate over networks.

Part 1: The Client-Server Model

✓ What is a Client?

- A **client** is a **device (like your PC or phone)** that **requests services or resources** from a server.
- Example: Your web browser (Chrome, Safari) is a **client** when it asks for a web page.

✓ What is a Server?

- A **server** is a **host (computer or system)** that **provides services or resources** to clients.
- Examples:
 - **Web server** → delivers websites
 - **Email server** → sends/receives emails
 - **File server** → stores and shares files

 **Clients and servers communicate using standard protocols** (rules) so they can understand each other — even if they're made by different companies.

Clients and Servers Over the Internet

- When you use apps like email, web browsers, or video streaming, you're almost always acting as a **client**, getting data from a **remote server**.
 - All these interactions rely on **agreed-upon protocols** at the **Application Layer (Layer 7)** of the networking model.
-

Part 2: URI, URL, and URN – How We Identify Web Resources

What is a URI?

- **URI = Uniform Resource Identifier**
- It's a string of characters that **identifies a network resource** (like a webpage or file).
- Examples: <https://www.example.com/page.html>, <mailto:someone@example.com>

URI has two main types:

1. **URL (Uniform Resource Locator)** → Tells you **where** the resource is located and **how** to access it (e.g., <http://>, <ftp://>)
2. **URN (Uniform Resource Name)** → Identifies the **name** of a resource, but not necessarily where it is located

Breakdown of a URL (Example: `https://www.example.com/author/book.html#page155`)

Part	Description	Example
Protocol/Scheme	How to access the resource (e.g., HTTPS, FTP)	<code>https://</code>
Hostname	The server's domain name	<code>www.example.com</code>
Path/File	The specific page or file on the server	<code>/author/book.html</code>
Fragment	A specific part of the page (like a bookmark)	<code>#page155</code>

Part 3: Common Application Layer Services & Protocols

1. Web Services (HTTP/HTTPS & HTML)

- **HTTP (HyperText Transfer Protocol)** → Used to **request and send web pages** (Port 80)
- **HTTPS (HTTP Secure)** → Secure version of HTTP (Port **443**) — encrypts data for safety
- **HTML (HyperText Markup Language)** → The language used to **format and display web pages** in your browser

How it works:

1. Your browser uses **DNS** to find the website's IP address
2. Then sends an **HTTP request** to the web server (Port 80 or 443)
3. The server sends back the **web page (HTML code)**, which your browser displays

2. File Transfer (FTP)

- **FTP (File Transfer Protocol)** → Used to **upload or download files** between computers
- **Ports used:**
 - **Port 21** → Control connection (commands)
 - **Port 20** → Data connection (actual file transfer)

 **Used for:** Sharing files, uploading websites, managing remote files

3. Remote Access (Telnet & SSH)

- **Telnet (Port 23)** → Lets you **remotely control a server** via text commands (like using a keyboard directly connected to it)
 -  **Not secure** (data is sent as plain text — can be intercepted!)
- **SSH (Secure Shell, Port 22)** → **Secure replacement for Telnet** (encrypts all data)

 **Best Practice:** Always use **SSH** instead of Telnet when possible

4. Email Services (SMTP, POP3, IMAP4)

Email involves **three main protocols**:

Protocol	Purpose	Port	Type
SMTP	Send emails from client to server, or between servers	25	Outgoing
POP3	Download emails from server to your device (usually deletes them from server)	110	Incoming
IMAP4	Access & manage emails on the server (keeps them on server)	143	Incoming

 Example:

You send an email using SMTP, it goes to your email server, and then to the recipient's server (also using SMTP). They access it using POP3 or IMAP4.

5. Text & Instant Messaging

- Instant messaging (IM) allows **real-time text chat** between users
- Common on social media, websites, and apps like WhatsApp, Teams, Facebook Messenger
- Clients and servers (or peer-to-peer) exchange messages instantly over the internet

6. Internet Phone Calls (VoIP)

- **VoIP = Voice over IP**
- Turns your **voice into digital data** and sends it over the internet using IP packets
- Examples: Skype, Zoom, Microsoft Teams, Cisco Webex
- Can call other VoIP users **or** connect to regular phones via a **gateway**

 **Requires:** Microphone, speakers (or headset), and VoIP software or app

Check Your Understanding – Key Takeaways

Concept	What It Means
Client-Server Model	Clients request services; servers provide them (e.g., your browser asks a web server for a page)
URI / URL / URN	Ways to name or locate resources on the web (like websites or files)
HTTP / HTTPS	Protocols for loading websites (HTTPS is secure)
FTP	Used to upload/download files between computers
Telnet / SSH	Remote access to servers (SSH is secure, Telnet is not)
SMTP / POP3 / IMAP4	Protocols for sending and receiving emails
VoIP	Making voice calls over the internet using digital data

Simple Summary Table

Service	Protocol(s)	Port(s)	Purpose
Web Browsing	HTTP, HTTPS	80, 443	View websites
File Transfer	FTP	20, 21	Upload/download files
Remote Access	Telnet (23), SSH (22)	23, 22	Control servers remotely
Email Sending	SMTP	25	Send emails
Email Receiving	POP3 (110), IMAP4 (143)	110, 143	Receive/manage emails
Instant Messaging	Varies	Varies	Chat in real-time
Voice Calls	VoIP (SIP, RTP, etc.)	Varies	Make calls over the internet

Module(17)

Introduction: Why This Topic Matters

sherif can't reach a website and realizes her computer has no internet connection. Madhav from IT arrives to help.

After Kishori explains she already tried pinging a website, Madhav does something different: he pings the default gateway.

He gets a reply, which tells him the **local network and router are working fine**. The problem must be with the **Internet Service Provider (ISP)**.

 How did he know to do that?

 He used basic troubleshooting commands to isolate the problem step-by-step. Let's learn how you can do it too!

Part 1: The ip config Command – Checking Your Own Settings

1. What is ip config?

- A **Windows command-line tool** that shows your computer's **IP configuration**.
- If your device has the **wrong IP address, subnet mask, or default gateway**, it won't be able to communicate on the network.
- ipconfig is the **first command you should use** to check your own settings.

✓ 2. Common ipconfig Commands

♦ ipconfig (Basic Info)

- Shows the most important information: **IPv4 Address, Subnet Mask, and Default Gateway**.

```
C:\> ipconfig
```

Wireless LAN adapter Wi-Fi:

```
IPv4 Address. . . . . : 10.10.10.130
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.10.10.1
```

♦ ipconfig /all (Detailed Info)

- Shows **everything**, including:
 - **MAC Address** (Physical Address)
 - **DNS Servers**
 - Whether **DHCP is enabled**
 - **DHCP Server** address
 - **Lease information** (how long you have the IP)

♦ ipconfig /release & ipconfig /renew (DHCP Refresh)

- If your IP information seems outdated or wrong, you can get a fresh one from the DHCP server.
- **ipconfig /release** → Tells the DHCP server you're **done with your current IP address**.
- **ipconfig /renew** → **Asks the DHCP server for a new IP address**.

🔧 This is a very common fix for network connectivity issues!

Part 2: The ping Command – Testing Connectivity

1. What is ping?

- The most common utility used to **test if you can reach another device** over the network.
 - You can ping an **IP address** or a **hostname** (like a website).
-

2. How Does ping Work?

- It sends a small packet called an **Echo Request** to the destination.
 - If the destination device receives it, it sends back an **Echo Reply**.
 - If you get a reply, it means **there is a connection** between your device and the destination.
-

3. How to Interpret Ping Results

♦ Successful Ping:

- You'll see "**Reply from...**" messages. This confirms a successful connection.
- It also shows the **round-trip time** in milliseconds (ms).

```
C:\> ping 10.10.10.1
```

```
Pinging 10.10.10.1 with 32 bytes of data:
```

```
Reply from 10.10.10.1: bytes=32 time=1ms TTL=64
```

```
Reply from 10.10.10.1: bytes=32 time=1ms TTL=64
```

♦ Failed Ping:

- You'll see messages like **"Request timed out"** or **"Destination host unreachable."**
 - This means there's a **connectivity problem** somewhere between you and the destination.
 - **Note:** Sometimes pings fail because a **firewall** is blocking them, not because the network is down.
-

♦ Pinging an IP vs. a Name (like `www.cisco.com`)

- When you ping a name, your computer first asks a **DNS server** to find its IP address.
 - **If pinging an IP works, but pinging a name fails**, you likely have a **DNS problem**.
-

♦ Pinging the Default Gateway

- This is a key troubleshooting step.
- **If you can ping your default gateway**, it means your **local network is working correctly**, and the problem is likely further out (like your ISP). This is exactly what Madhav did!
- **If you can't ping your default gateway**, the problem is on your **local network** (e.g., bad cable, switch issue, or router problem).

Check Your Understanding – Quick Review

Question 1:

You can't connect to the internet. What's the first command you should use to check your computer's IP address, subnet mask, and default gateway?

 Answer: ipconfig

Question 2:

You've confirmed your IP settings are correct. What command do you use to test if you can reach your router (default gateway)?

 Answer: ping <your_gateway_ip> (e.g., ping 10.10.10.1)

Question 3:

Your IP address seems wrong or outdated. Which two commands can you use to get a new IP address from the DHCP server?

 Answer: ipconfig /release followed by ipconfig /renew

Summary Table: Troubleshooting Commands

Command	What It Does
ipconfig	Shows your device's basic IP configuration (IP, subnet, gateway).
ipconfig /all	Shows detailed info, including MAC address and DNS servers.
ipconfig /release & /renew	Releases your current IP and requests a new one from the DHCP server.
ping	Tests connectivity to another device on the network by sending an echo request.

Do not forget us in your prayers